



FACULTY OF
COMPUTING
& INFORMATICS

FINAL YEAR PROJECT INTERIM REPORT

985

Machine Learning-Based System for Cardiopulmonary Sound
Separation

1221303805

AL-SALOUL, ASHRAF ALI HUSSEIN

BACHELOR OF COMPUTER SCIENCE B.CS (HONS) SOFTWARE
ENGINEERING

JULY 2026

985
Machine Learning-Based System for Cardiopulmonary
Sound Separation

BY

1221303805 AL-SALOUL, ASHRAF ALI HUSSEIN

PROJECT INTERIM REPORT SUBMITTED IN PARTIAL FULFILMENT OF THE
REQUIREMENT FOR THE DEGREE OF
BACHELOR OF COMPUTER SCIENCE B.CS (HONS) SOFTWARE ENGINEERING

in the

Faculty of Computing and Informatics

MULTIMEDIA UNIVERSITY
MALAYSIA

July 2026

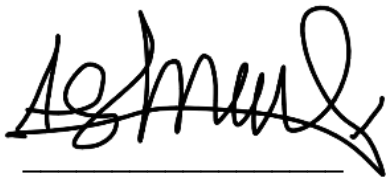
Copyright

© 2026 Universiti Telekom Sdn. Bhd. ALL RIGHTS RESERVED.

Copyright of this report belongs to Universiti Telekom Sdn. Bhd. as qualified by Regulation 7.2 (c) of the Multimedia University Intellectual Property and Commercialisation Policy. No part of this publication may be reproduced, stored in or introduced into a retrieval system, or transmitted in any form or by any means (electronic, mechanical, photocopying, recording, or otherwise), or for any purpose, without the express written permission of Universiti Telekom Sdn. Bhd. Due acknowledgement shall always be made of the use of any material contained in, or derived from, this report.

Declaration

I hereby declare that the work has been done by myself and no portion of the work contained in this report has been submitted in support of any application for any other degree or qualification of this or any other university or institution of learning.



Name of candidate: AL-SALOUL, ASHRAF ALI HUSSEIN

Faculty of Computing & Informatics

Multimedia University

Date: 07/07/2026

Acknowledgements

In the name of Allah, the Most Gracious, the Most Merciful.

I would like to express my sincere gratitude to my supervisor, Arshad Jamal, for his guidance, patience, and continuous support throughout the development of this project. His advice has been valuable in shaping both the technical and academic direction of this work.

I am also grateful to Multimedia University and the Faculty of Computing and Informatics for providing the academic environment, resources, and support necessary to complete this project.

Finally, I would like to thank my family, friends, and colleagues for their encouragement, patience, and moral support throughout this journey.

Abstract

Cardiopulmonary recordings may contain heart sounds, lung sounds, sensor-contact noise, body movement and environmental artefacts in the same audio channel. Because the heart and lung components can overlap in time and frequency, fixed filtering alone may not produce usable separated outputs. This application-based project proposes a Machine Learning-Based System for Cardiopulmonary Sound Separation. Its purpose is to provide a reusable prototype workflow that accepts a mixed cardiopulmonary recording and produces separate heart and lung sound files. The system is limited to sound separation and does not perform disease diagnosis or provide clinical decisions.

The FYP1 work combined a structured literature review, comparison of existing digital auscultation tools, requirements analysis, questionnaire data and software design. The literature review considered preprocessing, conventional decomposition methods, deep learning approaches, datasets and evaluation measures. A questionnaire collected 53 responses about processing challenges, preferred functions and interface expectations. Most respondents were from academic or technical backgrounds, so the results provide preliminary requirements rather than clinical validation. A targeted follow-up with healthcare staff and audio analysts is therefore required before the FYP2 requirements are finalised.

The revised design treats the project as an application instead of a stand-alone algorithm. Healthcare staff can enter non-identifying audio attributes, upload recordings, follow the job status and access outputs. An audio analyst can review the audio file list, select a separation model, run processing, inspect history and review valid evaluation information. Chapter 4 defines the application architecture, use cases, data records and separation strategies, including fixed-filter, NMF, VMD and NeoSSNet-style approaches. Chapter 5 limits itself to the activities planned for FYP2: implementation, integration, testing, deployment and documentation. No separation accuracy, user-acceptance result or clinical performance is claimed at this stage.

Table of Contents

Copyright	iii
Declaration	iv
Acknowledgements.....	v
Abstract	vi
Table of Contents.....	vii
List of Tables.....	x
List of Figures	xi
List of Abbreviations/Symbols.....	xii
List of Appendices	xiii
CHAPTER 1: INTRODUCTION	1
1.1 Overview	1
1.2 Problem Statement	1
1.3 Project Objectives	2
1.4 Project Scope.....	3
1.5 Project Limitations.....	3
1.6 Methodology.....	4
1.7 Target Audience	4
1.8 Summary.....	4
CHAPTER 2: LITERATURE REVIEW	6
2.1 Overview	6
2.2 Literature Search and Screening Process.....	6
2.3 Cardiopulmonary Sound Characteristics.....	7
2.4 Heart and Lung Sound Separation Techniques.....	8
2.4.1 Traditional Signal Processing Methods.....	8
2.4.2 Machine Learning and Deep Learning Methods.....	8
2.5 Datasets for Cardiopulmonary Sound Separation	9
2.6 Evaluation Metrics.....	9
2.7 Existing Systems and Prototype Relevance.....	10
2.7.1 Comparative Study of Existing Tools and Applications	10
2.7.2 Feature Decisions for the Proposed Solution	11

2.8 Literature Review Matrix	12
2.9 Research Gaps	13
2.10 Summary.....	13
CHAPTER 3: REQUIREMENTS ANALYSIS.....	14
3.1 Overview	14
3.2 Questionnaire.....	14
3.2.1 Questionnaire Design	14
3.2.2 Respondent Background and Experience	16
3.2.3 Question 1: Difficulty of Separating Heart and Lung Sounds.....	17
3.2.4 Question 2: Problems Affecting Cardiopulmonary Sound Processing	18
3.2.5 Question 3: Usefulness of a Heart and Lung Separation Prototype .	19
3.2.6 Question 4: Preferred System Features	20
3.2.7 Question 5: Preferred Output after Sound Separation	21
3.2.8 Question 6: Importance of Audio Preview Function	22
3.2.9 Question 7: Preferred Interface Style	23
3.2.10 Question 8: First Information Shown after Separation	24
3.2.11 Question 9: Importance of Evaluation Metrics	25
3.2.12 Question 10: Willingness to Use the Prototype.....	26
3.2.13 Target-User Validation Limitation and Follow-Up	27
3.3 Functional Requirements	28
3.4 Non-Functional Requirements	29
3.5 User Requirements	29
3.6 Summary.....	30
CHAPTER 4: SYSTEM DESIGN	31
4.1 Overview	31
4.2 Context Diagram	31
4.3 Use Case Diagram.....	32
4.3.1 Use Case Description	33
4.4 Activity Diagram	37
4.5 Class Diagram.....	38
4.5.1 Component-Level Architecture	39

4.5.2 Algorithm and Model Selection Design	41
4.5.3 Separation Algorithm Design and Mathematical Formulation	42
4.5.4 Database and Storage Design.....	46
4.6 Sequence Diagram	47
4.7 Interface Design	48
4.8 Summary.....	49
CHAPTER 5: IMPLEMENTATION PLAN	50
5.1 Development Phase	50
5.1.1 Front-End Development	50
5.1.2 Back-End Development.....	51
5.1.3 Machine Learning Model Integration	51
5.1.4 Database and Storage Integration.....	52
5.2 Testing.....	52
5.2.1 Unit Testing.....	53
5.2.2 Integration Testing	53
5.2.3 System Testing	53
5.2.4 User Acceptance Testing	54
5.3 Deployment.....	54
5.4 Documentation and Maintenance Plan	55
5.5 Summary.....	55
CHAPTER 6: CONCLUSION.....	56
6.1 Overview	56
6.2 Project Summary.....	56
6.3 Expected Contributions	56
6.4 Limitations.....	56
6.5 Future Work	57
6.6 Summary.....	57
References	58
Appendix A: Gantt Chart	64
Appendix B: FYP1 Meeting Logs.....	65
Appendix C: Turnitin Similarity Index Page.....	68
Appendix D: Full Literature Review Matrix.....	73

Appendix E: PRISMA Screening Summary	80
Appendix F: System Design Diagrams	81
Appendix G: Planned Test Cases and Evaluation Metrics	82
Appendix H: Full User Requirements Questionnaire	83

List of Tables

Table 2.1: Comparative study of cardiopulmonary sound tools and applications	10
Table 2.2: Feature decisions for the proposed cardiopulmonary sound separation solution	11
Table 2.3: Summary literature review matrix.....	12
Table 3.1: Questionnaire structure for the user requirements analysis	15
Table 3.2: Questionnaire analysis mapping	16
Table 3.3: Functional requirements for cardiopulmonary sound separation prototype.....	28
Table 3.4: Non-functional requirements for cardiopulmonary sound separation prototype.....	29
Table 3.5: User requirements for proposed prototype.....	30
Table 4.1: Use case description for UC01 - Enter audio file attributes	34
Table 4.2: Use case description for UC02 - Upload mixed cardiopulmonary audio.....	34
Table 4.3: Use case description for UC03 - View audio file list	34
Table 4.4: Use case description for UC04 - Select separation model	35
Table 4.5: Use case description for UC05 - Run sound separation	35
Table 4.6: Use case description for UC06 - View processing status	35
Table 4.7: Use case description for UC07 - Preview separated outputs	35
Table 4.8: Use case description for UC08 - Download separated outputs	36
Table 4.9: Use case description for UC09 - View processing history	36
Table 4.10: Use case description for UC10 - View evaluation metrics	36
Table 4.11: Main system modules and responsibilities	40
Table 4.12: Options for model selection in the cardiopulmonary sound separation prototype	41
Table 4.13: Database entities for prototype traceability and analysis	46
Table 4.14: Interface areas for cardiopulmonary sound separation	49
Table 5.1: Planned backend API responsibilities	51

List of Figures

Figure 2.1: PRISMA flow diagram.....	7
Figure 3.1: Perceived difficulty of separating heart and lung sounds.....	18
Figure 3.2: Problems affecting cardiopulmonary sound processing.....	19
Figure 3.3: Usefulness of the proposed heart and lung separation prototype...	20
Figure 3.4: Preferred system features	21
Figure 3.5: Preferred output after sound separation	22
Figure 3.6: Preview separated heart and lung sounds.....	23
Figure 3.7: Preferred interface style	24
Figure 3.8: Preferred first information after separation is completed	25
Figure 3.9: Importance of evaluation metrics for separated output.....	26
Figure 3.10: Willingness to use the cardiopulmonary sound separation prototype	27
Figure 4.1: Context diagram for the proposed cardiopulmonary sound separation prototype	32
Figure 4.2: Use case diagram for the cardiopulmonary sound separation prototype.....	33
Figure 4.3: Activity diagram for the sound separation workflow	38
Figure 4.4: Class diagram for the cardiopulmonary sound separation prototype	39
Figure 4.5: Component-level architecture for the cardiopulmonary sound separation prototype	40
Figure 4.6: Sequence diagram for model selection and cardiopulmonary sound separation.....	48

List of Abbreviations/Symbols

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
DL	Deep Learning
EPS	Small epsilon value used to avoid division by zero
FFT	Fast Fourier Transform
FYP	Final Year Project
HLS-CMDS	Heart and Lung Sounds Dataset Recorded from a Clinical Manikin using Digital Stethoscope
HTTP	Hypertext Transfer Protocol
ID	Identifier
ML	Machine Learning
NeoSSNet	Deep learning model for neonatal chest sound separation
NMF	Non-negative Matrix Factorization
PCG	Phonocardiogram
PESQ	Perceptual Evaluation of Speech Quality
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
S1	First heart sound
S2	Second heart sound
SAR	Signal-to-Artifacts Ratio
SDR	Signal-to-Distortion Ratio
SIR	Signal-to-Interference Ratio
SI-SDR	Scale-Invariant Signal-to-Distortion Ratio
SNR	Signal-to-Noise Ratio
SQLite	Lightweight relational database engine
STFT	Short-Time Fourier Transform
STOI	Short-Time Objective Intelligibility
UI	User Interface
UML	Unified Modeling Language
VMD	Variational Mode Decomposition
WAV	Waveform Audio File Format

List of Appendices

<u>Appendix A: Gantt Chart</u>	64
<u>Appendix B: FYP1 Meeting Logs</u>	65
<u>Appendix C: Turnitin Similarity Index Page</u>	68
<u>Appendix D: Full Literature Review Matrix</u>	73
<u>Appendix E: PRISMA Screening Summary</u>	80
<u>Appendix F: System Design Diagrams</u>	81
<u>Appendix G: Planned Test Cases and Evaluation Metrics</u>	82
<u>Appendix H: Full User Requirements Questionnaire</u>	83

CHAPTER 1: INTRODUCTION

1.1 Overview

Cardiopulmonary auscultation captures sounds produced by the heart, lungs, airways, the chest wall and the recording environment. Typical recordings contain useful information about the cardiovascular and respiratory systems masked by artifacts such as breathing sounds, body movement, motion, sensor noise and room noise. This renders a practical software design challenge since the user is not interested in a mixture but rather another recording without the unwanted contribution. This project will focus on the design of a prototype workflow that can be reused to separate cardiopulmonary sounds.

The project's name is Machine Learning Based System for Cardiopulmonary Sound Separation. This is an application based FYP1 project and the contribution is prototype/proof of concept and NOT a medical product. This stage is to prepare the foundation for the FYP2 which consists of background study, literature review, requirements analysis, system design and implementation plan.

1.2 Problem Statement

1. Noise in a recording, plus additional artefacts generated during the recording, will degrade the quality of your Cardiopulmonary Sound Recordings before even attempting to process them with any software. In practice, sensor noises, artefacts from breathing movements, unwanted movement artefacts, background noises, and limitations of recordable quantities by the recording devices, can contaminate heart and lung sounds. The most important limitation here is that this is a serious concern to all of our separation tasks, because it means that the model will be fed with a distorted signal as input instead of the ideal clean signal. Research works regarding noisy neonatal chest sounds, blind denoising, and lung sound enhancement have demonstrated that preprocessing is an essential step before attempting further downstream analysis or separation ([Grooby et al., 2023](#); [Fattahi et al., 2022](#); [Ali et al., 2023](#)). This means that for our FYP, the choice of model is not the only consideration. Your software will also require file validation and additional preprocessing before the actual separation can take place.

2. Heart and lung sounds are difficult to differentiate because they may co-occur both in time and frequency. Short heart sound events and longer respiratory patterns can be simultaneously recorded on a cardiopulmonary recording. These frequency ranges can overlap, and a fixed frequency filter could remove useful information or retain unwanted sound in the separated output. With only one mixed recording and no isolated source signals, separation becomes more difficult because the system has limited information about the original heart and lung components. Recent work on neonatal heart and lung sound separation, decomposition-based methods, and compact learning models suggests that this task needs more than a simple filtering rule (Poh et al., 2024; Sun et al., 2024; Wang et al., 2023a). This project responds to that gap by planning a machine learning-based separation workflow that can support different separation strategies during FYP2.
3. A third difficulty is that there is no clear reusable software workflow for students/researchers interested in using cardiopulmonary sound separation as an application. Many related studies focus on diagnosis, classification, or algorithm-level experiments. The above studies are useful but not always illustrative of the steps a user would take in uploading one mixed recording and validating it, preprocessing it, choosing a model, running separation, and downloading the separated heart and lung outputs. Review and analysis papers also reveal that much of the research in the field of cardiopulmonary sounds focuses on the clinical interpretation or the model performance instead of the surrounding prototype design (Roy et al., 2024; Zhao et al., 2024; Torabi et al., 2025a). Since this is a Software Engineering FYP, the system design matters as much as the selected algorithm. The prototype therefore needs clear requirements, defined module responsibilities, an interface workflow, a storage plan and an evaluation plan so that it can be implemented systematically during FYP2.

1.3 Project Objectives

1. To identify and apply relevant preprocessing techniques to reduce noise and enhance the quality of the cardiopulmonary recordings.
2. To propose a machine learning-based method that can separate mixed cardiopulmonary recordings into heart and lung sound outputs.
3. To develop a prototype, reusable in research, and evaluate the developed tool using appropriate public datasets and performance metrics.

1.4 Project Scope

The current project scope is a prototype for cardiopulmonary sound separation. The application is designed around a practical clinic- or laboratory-oriented workflow: healthcare staff enter non-identifying recording attributes and upload one mixed cardiopulmonary audio file, while an audio analyst reviews the file record, selects a separation strategy, starts processing and inspects the generated outputs. The system validates and preprocesses the input before generating separate heart and lung sound files for preview and download.

As per the project's scope, this work will involve publicly available or otherwise approved datasets, preprocessing, model integration, and evaluation planning with relevant performance metrics. This prototype is not intended to detect, diagnose, or recommend any medical treatment or procedure. It is also not designed to operate in a clinical environment or replace the professional judgment of a medical practitioner. Instead, it will serve as a research and educational prototype for separation tasks only.

1.5 Project Limitations

The project's scope is limited to the FYP1 stage; thus, the current report will focus on background, requirements, design, and planning. The actual model training, implementation, user acceptance, and performance assessment will be carried out in the FYP2 stage. Similarly, the proposed prototype is limited by the availability of public or otherwise approved cardiopulmonary recordings. The data gathered from the questionnaires are based on the convenience sample of students, and thus they only represent general requirements. This limitation is standard for FYP1, but it is essential to acknowledge it when elaborating on the findings.

1.6 Methodology

This work will follow the Application-Based methodology that involves investigation, definition, design, and planning phases. First, the project will review the characteristics of cardiopulmonary recordings, potential separation procedures, public datasets, and performance metrics. Second, the requirements will be established by analyzing the literature, existing solutions or systems, supervisor's notes, and the requirements survey results. Third, the design will be created with the support of multiple artefacts, including Context, Use Case, and Activity Diagrams for the process and the Class and Sequence Diagrams for implementation specifics. Finally, the FYP1 will conclude with the planning of Front End, Back End, machine learning integration, storage, testing, deployment, documentation, and maintenance stages in FYP2.

This methodology will help this project to follow the clear logic of the Application-Based task. It will also support the report's claims by separating requirements, design, and implementation phases. In particular, the FYP1 report will detail the findings, but it will avoid presenting results that will only be available in the FYP2 stage.

1.7 Target Audience

The primary application roles are healthcare staff and audio analysts. Healthcare staff represent the person who prepares a recording, enters non-identifying audio attributes and uploads the file. The audio analyst represents the person who reviews the audio list, selects a separation model, runs processing and examines the separated outputs. Students and researchers remain secondary users for development and evaluation, but the workflow is no longer described as being for any unspecified user.

Furthermore, the project will benefit the student developer by structuring the FYP2 work and outlining the tasks of input file verification, preprocessing, model training, cardiopulmonary separation, storage, preview, download, and performance evaluation.

1.8 Summary

This chapter described the project's background, scope, limitations, methodology, and audience. The project aims to develop a prototype of cardiopulmonary separation

software. The primary challenge is to ensure that the cardiopulmonary recordings will remain consistent in terms of quality and free from noise and interference. It was found that many works report cardiopulmonary separation, but the focus of this project is to provide a complete prototype for the separation task. The next chapter will provide details on the relevant literature and research findings.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview

The review focuses on the studies that served as the foundation for developing the cardiopulmonary sound separation prototype. Specifically, the review dwells on sound characteristics, separation methods, datasets, evaluation criteria, existing systems, and the relevance of prototypes to cardiopulmonary separation. Thus, the review's objective is to furnish a basis for the cardiopulmonary prototype in terms of data preprocessing, model-based processing, outputs, and evaluation.

The common problem with the prototypes described in the reviewed articles is the presence of cardiac, pulmonary, and recording artifacts in the same channel. The conventional signal processing approaches can partly address this issue, as they served as the foundation for building more complex models. Nevertheless, the current state-of-the-art in machine learning and deep learning seems to be more relevant to the task at hand. The problem of sound separation is more involved than linear signal processing and benefits from the flexibility of neural networks.

2.2 Literature Search and Screening Process

The search strategy comprised cardiopulmonary separation methods, including cardiopulmonary sound separation, heart sound removal algorithms, lung sound separation, and denoising. The strategy identified the relevant literature on public auscultation databases, evaluation metrics, and cardiopulmonary sound prototype systems. Initially, 96 records were retrieved; however, only 49 were included in the final screening process. Consequently, 35 articles were selected for inclusion in this review and the subsequent creation of a literature matrix for the full systematic review. In total, 18 articles were included in the main topic of cardiopulmonary separation methods, and 17 constituted the background/supplementary category. Finally, 14 records were excluded from the current study but are available for future research.

The review follows the PRISMA methodology, and Figure 2.1 below summarizes the literature search and screening process.

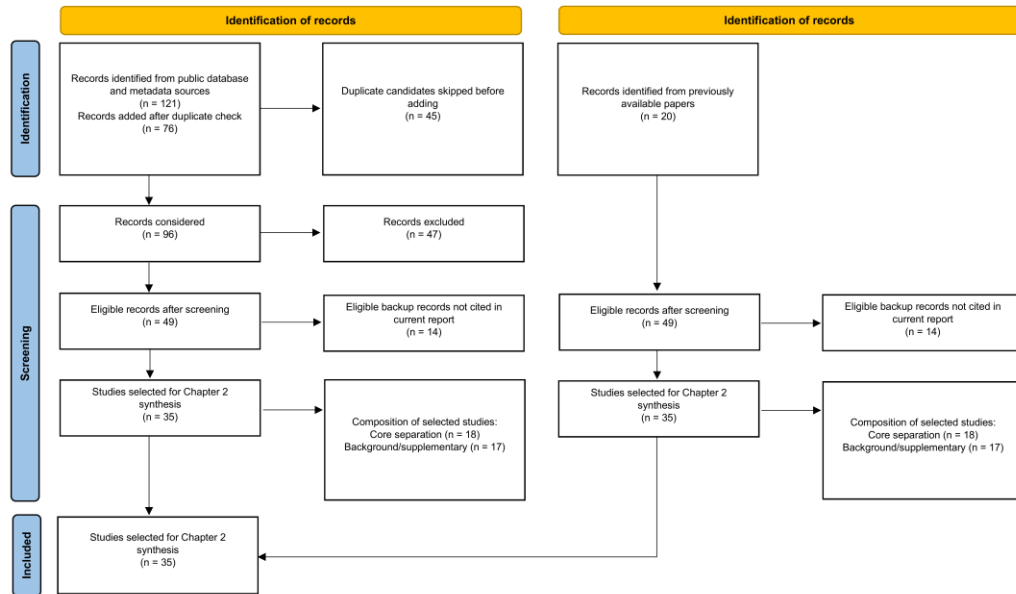


Figure 2.1: PRISMA flow diagram

2.3 Cardiopulmonary Sound Characteristics

Recordings of cardiopulmonary sounds contain a combination of cardiac, pulmonary, chest wall, and recording artifacts vibrations. Heart activity is typically present in the form of periodically occurring S1 and S2 clicks. Lung sounds are associated with the inhalation-exhalation process and vary depending on the stage of breathing. Since both heart and lung activity are recorded from the same surface, their spectra may overlap, making separation challenging. This effect is particularly noticeable when using recordings from a single channel.

The phenomenon is explained in detail in the studies on respiratory sound analysis and cardiopulmonary sound characteristics ([Roy et al., 2024](#); [Zhao et al., 2024](#); [Grooby et al., 2023](#)). According to these records, a cardiopulmonary sound separation prototype must address the interference from multiple sources and varying levels of noise. Hence, the necessity of preprocessing and standardized input preparation for model-based separation.

2.4 Heart and Lung Sound Separation Techniques

Heart and lung sound separation is a well-researched topic that includes traditional signal processing approaches and machine learning methods. In particular, cardiopulmonary separation research shines a light on the significance of selecting separation methods based on specific conditions, such as recording sources, noise levels, training data availability, and evaluation criteria.

2.4.1 Traditional Signal Processing Methods

Traditional signal processing methods include filtering, adaptive filtering, blind source separation, non-negative matrix factorization, empirical mode decomposition, variational mode decomposition, and similar techniques. The primary benefit of such an approach is its explainability, which is critical for a cardiopulmonary sound separation prototype. Moreover, many signal processing methods can be applied without extensive training data. Blind denoising and decomposition-based separation are directly relevant to cardiopulmonary separation, as they tackle the issues of mixed heart and lung recordings ([Fattahi et al., 2022](#); [Sun et al., 2024](#); [Ullah & Zhang, 2024](#)).

However, the limitations of traditional methods are also considerable, particularly the inability to manage complex or varying sources of interference. Hence, a method that works for one set of conditions may fail for another recording. For these reasons, traditional signal processing methods can be used as a baseline or as part of a prototype's preprocessing stage.

2.4.2 Machine Learning and Deep Learning Methods

Machine learning and deep learning approaches to cardiopulmonary separation learn the patterns that can distinguish between heart and lung sounds. Recent research points out the effectiveness of neural networks, source separation algorithms, compact models, and data-driven methods for cardiopulmonary sound analysis ([Poh et al., 2024](#); [Wang et al., 2023a](#); [Zhang et al., 2025](#); [Ali et al., 2023](#)). These methods are directly relevant to the current project because they can be trained to process a single input and separate heart and lung sounds.

The main advantage of machine learning approaches is their ability to capture complex patterns in data, including those that cannot be described by fixed rules. The limitation

of such methods is their dependence on training data and the computational power of the underlying hardware. Hence, the cardiopulmonary separation model must be trained on a suitable dataset and have consistent preprocessing and evaluation pipelines. Finally, the choice of a particular algorithm is not critical to the current research since the primary focus is on the prototype as a whole.

2.5 Datasets for Cardiopulmonary Sound Separation

Public datasets are crucial for cardiopulmonary sound separation because they enable consistent evaluation across different studies. The review includes articles that describe datasets of heart, lung, and cardiopulmonary sounds. The studies are relevant to the current research because they provide the basis for preprocessing, model training, benchmarking, and evaluation ([Torabi et al., 2025a](#); [Grooby et al., 2022](#); [Ali et al., 2026](#); [Rocha et al., 2023](#)).

For the current project, it is critical to select the appropriate dataset for the intended application. Ideally, the dataset should include heart and lung sounds or allow the construction of mixed recordings with consistent labels. At the same time, a dataset with restricted or unavailable access is not an option for the current FYP, as the project timeline does not allow for extensive data collection and preparation efforts.

2.6 Evaluation Metrics

A cardiopulmonary sound separation prototype is not something that can be assessed visually, and therefore one has to look beyond the waveform to determine whether the prototype is any good. One might argue that an inspection of the waveform would suffice, but it is recommended that more objective criteria be used for this type of prototype. For example, one may want to check whether the model is able to produce results close to the expected output without interference, and how applicable the prototype is. Processing speeds and the prototype's resource intensity are also important considerations, as the cardiopulmonary separation prototype is meant for student/research purposes.

Finally, it is recommended that performance, source separation, and quality be evaluated to ensure that the prototype actually accomplishes its intended functions. In particular, the review highlights the importance of objective metrics for heart and lung separation

([Torabi et al., 2025a](#); [Roy et al., 2024](#); [Zhao et al., 2024](#)). For the current project, the review serves as the basis for selecting appropriate metrics for the selected dataset and evaluation pipeline.

2.7 Existing Systems and Prototype Relevance

Existing work includes research platforms and commercial digital-auscultation applications. Their main value to this FYP is not that they already separate mixed heart and lung sounds, but that they show how body-sound recordings are captured, described, organised, reviewed and shared in a usable application. Eko supports recording, playback, annotation, waveform or PCG viewing, storage and sharing functions, while the Littmann CORE device uses the Eko application for visualisation and recording support (3M, n.d.; Eko Health, 2025). Thinklabs Wave provides real-time waveform and frequency displays together with recording, annotation, file browsing and sharing features (Thinklabs Medical LLC, n.d.). StethAid extends this idea into a platform with a digital stethoscope, mobile applications, web portals and centralised storage (Arjoun et al., 2023).

These applications demonstrate that a useful solution needs more than a model execution button. Recording attributes, a clear audio-file list, traceable processing status, result review and controlled output handling are part of the application workflow. The proposed FYP does not copy diagnostic or patient-management functions from these products. Instead, it adopts the relevant software features and adds the project-specific contribution: selectable heart-lung sound separation strategies that produce separate audio outputs.

2.7.1 Comparative Study of Existing Tools and Applications

The following comparison identifies the application features that are relevant to the proposed prototype and the gap that remains for this project.

Table 2.1: Comparative study of cardiopulmonary sound tools and applications

Tool or application	Main workflow	Observed features	Relevance and limitation for this FYP
Eko App	Digital auscultation and recording review	Record, play back, annotate and save body sounds; display PCG	Supports preview, notes, visualisation and record management. It does not

Tool or application	Main workflow	Observed features	Relevance and limitation for this FYP
		waveforms; store and share recordings (Eko Health, 2025).	provide the proposed selectable heart-lung separation workflow.
3M Littmann CORE with Eko App	Device-assisted digital auscultation	Visualise, record, save, annotate and share sounds through the connected application (3M, n.d.).	Shows a practical healthcare recording workflow. Device integration and diagnosis functions remain outside this FYP scope.
Thinklabs Wave	Capture and review digital body sounds	Waveform and frequency display, recording, annotation, file browsing, storage and sharing (Thinklabs Medical LLC, n.d.).	Supports the need for audio lists, attributes, preview and optional waveform or spectrogram displays.
StethAid	Pediatric digital-auscultation platform	Digital stethoscope, mobile applications, user portals, sound libraries and centralised storage (Arjoun et al., 2023).	Shows how roles, records and storage form an application. Its clinical analysis functions are not adopted by this separation-only prototype.
Proposed FYP solution	Role-based upload and heart-lung sound separation	Audio attributes, file validation, audio list, model selection, processing status, separate outputs, preview, download, history and valid metrics.	Adds an end-to-end, model-selectable separation workflow while avoiding diagnosis claims.

2.7.2 Feature Decisions for the Proposed Solution

The comparison was translated into a feature decision so that the proposed system remains achievable as an FYP prototype. Core features support the complete application flow. Additional features can be developed after that flow is stable, while diagnosis and treatment functions are excluded.

Table 2.2: Feature decisions for the proposed cardiopulmonary sound separation solution

Feature	Decision	Reason
Audio attributes and upload	Core FYP2	Provides context for each recording and supports a healthcare-staff workflow.
File validation and correction messages	Core FYP2	Prevents unsupported or unreadable audio from entering the separation process.
Audio file list and filters	Core FYP2	Allows an audio analyst to find records and review their processing state.
Model selection and separation	Core FYP2	Implements the main project contribution and supports strategy comparison.
Status, preview, download and history	Core FYP2	Completes the user workflow and makes outputs traceable.
Waveform or spectrogram display	Additional FYP2	Improves review but is secondary to reliable output generation.
Notes, annotations and audit information	Additional FYP2	Supports handover between roles after the core workflow is stable.
Evaluation metrics	Conditional	Display only when reference data or a valid evaluation procedure is available.

Feature	Decision	Reason
Disease diagnosis or treatment advice	Excluded	Outside the approved sound-separation scope and not supported by this prototype.

2.8 Literature Review Matrix

Table 2.3 below provides a brief overview of the studies that were considered for the current research. In particular, the matrix below shows how each study was used in the preparation of the current report. The full literature review matrix is available in Appendix D.

Table 2.3: Summary literature review matrix

Study	Focus	Method or contribution	Relevance to this project
Grooby et al. (2023)	Noisy neonatal chest sounds	Examines the qualities of noisy recordings and processing requirements	Supports the importance of considering preprocessing and input preparation
Fattahi et al. (2022)	Blind denoising	Denoising strategy for respiratory or cardiopulmonary recordings	Supports the preparation-focused approach and separation prerequisites
Poh et al. (2024)	Neonatal separation	ML-based heart and lung separation	Supports the direction of separation
Sun et al. (2024)	Decomposition-based separation	Separation strategy based on NMF and VMD analysis	Provides a traditional approach baseline direction
Ullah and Zhang (2024)	Unsupervised separation	Unsupervised heart and lung separation approach	Supports the consideration of alternatives
Wang et al. (2023a)	Compact learning model	Compact model for cardiopulmonary processing	Supports the prototyping-focused model
Zhang et al. (2025)	Joint separation approach	DL-based sound separation architecture	Supports the ML-based separation development
Torabi et al. (2025a)	Dataset and benchmarking	Cardiopulmonary sounds dataset / benchmark support	Supports the dataset and evaluation considerations
Roy et al. (2024)	Review study	Review of processing and analysis considerations	Supports the problem framing and research gap analysis
Zhao et al. (2024)	Heart sound analysis	Review or analysis of heart sound processing	Supports the evaluation and relevance analysis

2.9 Research Gaps

Across the reviewed studies, three key gaps were identified. First, cardiopulmonary recordings are notoriously noisy, and mixed heart-lung recordings include artifacts from the recording equipment and the subject's breathing. Second, heart and lung components may overlap, reducing the usefulness of any single-channel recordings. Third, most of the reviewed studies were focused on diagnosis or classification, not on an end-to-end prototype. These gaps are critical to the current research since the cardiopulmonary sound separation prototype will be applied to real-world data with mixed sources and limited training data.

2.10 Summary

This review connected cardiopulmonary sound research with the design of an application-based solution. The technical studies support noise-aware preprocessing, conventional baselines, machine learning strategies, public datasets and objective evaluation. The comparison of Eko, Littmann CORE, Thinklabs Wave and StethAid added an application perspective: useful systems organise recordings, preserve attributes, provide file lists, show progress, support review and keep outputs traceable. These findings directly inform the role-based requirements in Chapter 3 and the system design in Chapter 4.

CHAPTER 3: REQUIREMENTS ANALYSIS

3.1 Overview

The requirements for the Machine Learning-Based System for Cardiopulmonary Sound Separation were identified from the project scope, literature and document review, comparison of existing applications, supervisor consultation and a questionnaire. The Google Form received 53 responses and provided useful early information about processing challenges, preferred features and interface expectations. However, most respondents came from academic or technical backgrounds rather than hospital or clinic roles. The results are therefore treated as preliminary requirements, and a targeted follow-up with healthcare staff and audio analysts is required before the FYP2 requirements are finalised.

3.2 Questionnaire

The questionnaire was used to collect practical preferences for the prototype. It covered respondent background, cardiopulmonary processing challenges, system features, interface style, output presentation and evaluation needs. Ten requirement-focused items are analysed in this chapter, while the complete 24-question instrument remains in Appendix H.

The full questionnaire contains 24 items and is provided in Appendix H. Chapter 3 analyses 10 requirement-focused items to avoid repeating closely related questions while preserving traceability between the questionnaire and the resulting requirements.

3.2.1 Questionnaire Design

The questionnaire was distributed through Google Forms and received 53 responses. The initial target included Software Engineering, Computer Science, artificial intelligence, machine learning, biomedical engineering and health-related students, researchers and supervisors. This convenience sample was suitable for early prototype feedback, but it did not adequately represent the

healthcare staff and audio analysts expected to use a clinic- or laboratory-oriented application.

The introduction explained the academic purpose of the study, confidentiality and the absence of right or wrong answers. Multiple-choice, checkbox, Likert-scale and paragraph questions were used. The collected responses support preliminary feature and usability decisions. They do not constitute clinical validation or proof that the prototype is ready for hospital use.

The structure of the questionnaire is given in Table 3.1 below.

Table 3.1: Questionnaire structure for the user requirements analysis

Section	Purpose	Example question focus	Question type
Section A: Respondent Background	Identify the respondent's background and level of expertise in the field.	Age group, study background, signal processing experience, biomedical audio background, demographic data, and field experience.	Multiple choice
Section B: Current Challenges in Cardiopulmonary Audio Processing	Identify the cardiopulmonary audio processing issues that respondents may encounter.	Separation difficulties, noise, source overlap, recording quality, and absence of simple tools.	Multiple choice, checkboxes, Likert-style scale
Section C: System Feature Preferences	Highlight the prototype features that users may prefer for cardiopulmonary audio processing.	Upload, preprocessing, separation, preview, download, history, and evaluation features.	Multiple choice, Likert-style scale
Section D: Interface and Output Preferences	Identify the interface displays and cardiopulmonary audio result presentations that users may find useful and clear.	Audio player, waveform display, spectrogram display, metrics, interface simplicity, and error messages.	Multiple choice, Likert-style scale
Section E: Evaluation and Improvement	Identify what users expect to evaluate and what	Metrics, processing information, willingness to use the prototype, and	Multiple choice, Likert-style scale,

	improvements they may suggest.	improvement suggestions.	paragraph response
--	--------------------------------	--------------------------	--------------------

Table 3.2 below provides an overview of the questionnaire sections and shows how they were mapped to the 10 items used to structure the analysis in this chapter.

Table 3.2: Questionnaire analysis mapping

Report item	Main chart used	Related questions used only for discussion	Requirement area supported
Respondent Background and Experience	Section A, Q2-Q5	Section A, Q1 when needed	Respondent profile
Difficulty of Separating Heart and Lung Sounds	Section B, Q1	None	Separation requirement
Problems Affecting Cardiopulmonary Sound Processing	Section B, Q2	Section B, Q3	Preprocessing and separation requirements
Usefulness of Prototype	Section B, Q4	None	System relevance
Preferred System Features	Section C, Q1	Section C, Q6	Functional requirements
Preferred Output After Separation	Section C, Q2	Section D, Q1	Output design
Importance of Audio Preview Function	Section C, Q3	Section C, Q4	Preview and download requirements
Preferred Interface Style	Section D, Q2	Section D, Q3 and Q5	Usability requirements
First Information Shown After Separation	Section D, Q4	Section C, Q5 and Q6	Result page design
Importance of Evaluation Metrics	Section E, Q1	Section E, Q2	Evaluation requirements
Willingness to Use Prototype	Section E, Q3	Section E, Q4	User acceptance and improvements

3.2.2 Respondent Background and Experience

Section A collected respondent background information, including age group, respondent category, familiarity with audio or signal processing, biomedical-audio experience and experience with audio-related tasks. These responses describe

the sample, so they are summarised in text rather than presented as separate figures.

From the 53 respondents, 39 or 73.6% of the respondents selected the 18-25 years age group, 8 or 15.1% selected the below 18 years age group, and the remaining 6 or 11.3% selected the 26-35 years age group. In terms of the respondent category, 20 or 37.7% respondents identified as Software Engineering or Computer Science students, 15 or 28.3% respondents selected AI or Machine Learning students, 8 or 15.1% selected Biomedical Engineering or related health students, and 7 or 13.2% selected researchers. The remaining respondents selected from other categories including lecturer/supervisor and other respondents.

With respect to familiarity with audio or signal processing, 29 respondents (54.7%) selected basic, 15 (28.3%) selected not familiar, 6 (11.3%) selected intermediate and 3 (5.7%) selected advanced. Thirty respondents (56.6%) reported prior biomedical-audio experience, 22 (41.5%) reported no prior experience and 1 (1.9%) was unsure. The sample therefore provides useful feedback from technical and student users, including some health-related respondents, but it cannot be treated as representative of hospital personnel. That limitation affects how confidently the requirements can be generalised to a clinic workflow.

3.2.3 Question 1: Difficulty of Separating Heart and Lung Sounds

The first research question focused on asking the respondents to evaluate the level of difficulty of separating heart and lung sounds from the mixed cardiopulmonary

recordings. The question aimed to establish whether the cardiopulmonary sounds separation task qualifies as a technical challenge.

1. How difficult do you think it is to separate heart and lung sounds from a mixed cardiopulmonary recording?
53 responses

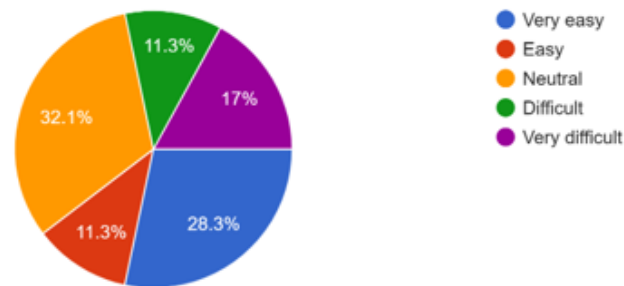


Figure 3.1: Perceived difficulty of separating heart and lung sounds

Finding: Responses were mixed. The largest group, 17 respondents (32.1%), selected neutral. A total of 21 respondents (39.6%) selected easy or very easy, while 15 (28.3%) selected difficult or very difficult. These results do not show a clear agreement that separation is either simple or difficult from the user's perspective. They do, however, support keeping the technical separation process behind a straightforward application workflow.

Requirement implication: The interface should provide a streamlined process that allows users to upload a mixed cardiopulmonary recording and receive separate heart and lung outputs without requiring them to configure the internal signal-processing steps.

3.2.4 Question 2: Problems Affecting Cardiopulmonary Sound Processing

The second research question asked the respondents to identify the main problems affecting cardiopulmonary recording or processing. The options included overlapping of heart and lung sounds, challenges in identifying useful sound components, background noises, low recording quality, poor visual representation of the results, and fewer available software tools.

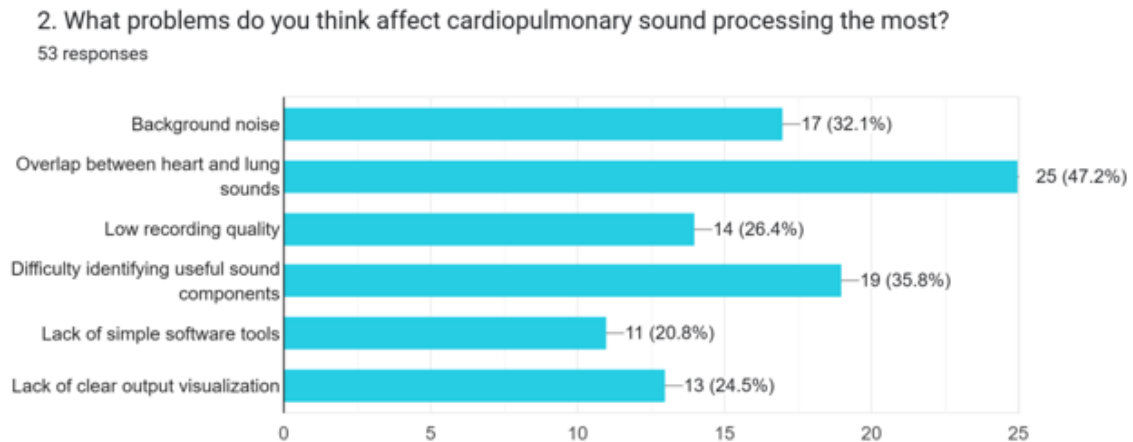


Figure 3.2: Problems affecting cardiopulmonary sound processing

Finding: This question allowed multiple responses, which is evident from the relatively high percentage of respondents that selected overlapping of heart and lung sounds (25 or 47.2%). Furthermore, many respondents indicated that the inability to easily identify useful sound components (19 or 35.8%), background noises (17 or 32.1%), low recording quality (14 or 26.4%), and the need for better output visualisation (13 or 24.5%) were major problems affecting cardiopulmonary recordings. Similarly, fewer available software tools for cardiopulmonary analysis were also a problem according to 11 or 20.8% of the respondents. In the related noisy audio question, 27 or 50.9% of the respondents either strongly agreed or agreed that noisy audio recordings could affect the accuracy of machine learning-based sound analysis.

Requirement implication: The cardiopulmonary separation prototype needs built-in audio validation, preprocessing, and separation features to address overlapping of heart and lung sounds, background noises, and low-quality audio recordings. The evidence further indicates the need for visualisation features to help the user easily inspect the separated cardiopulmonary recordings.

3.2.5 Question 3: Usefulness of a Heart and Lung Separation Prototype

The third research question asked the respondents whether a simple heart and lung separation prototype would be useful to students, researchers, or users of biomedical audio data.

4. Do you think a simple software prototype for heart-lung sound separation would be useful for students or researchers?

53 responses

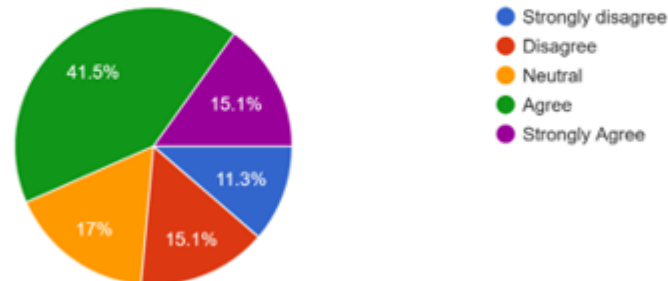


Figure 3.3: Usefulness of the proposed heart and lung separation prototype

Finding: Twenty-two respondents (41.5%) selected agree and 8 (15.1%) selected strongly agree. Together, 30 respondents (56.6%) gave a positive response. The remaining 23 respondents (43.4%) selected neutral, disagree or strongly disagree. The results therefore show more positive than non-positive responses, while still leaving a substantial group that was neutral or unconvinced.

Requirement implication: This finding implies that the cardiopulmonary separation prototype should focus on addressing the cardiopulmonary analysis challenge. As such, it would be useful to focus on an application-oriented design and provide a simple prototype that allows users to upload a mixed cardiopulmonary audio, separate the audio recording into heart and lung sounds, preview the separated sounds, and download the audio files.

3.2.6 Question 4: Preferred System Features

The fourth research question aimed to identify the system features that would be preferred by the system users. The system features included uploading a mixed cardiopulmonary audio, preprocessing of noisy recordings, separating the cardiopulmonary audio into heart and lung sounds, preview separated audio, download separated audio, view processing history, and view evaluation results.

1. Which feature would be most important in the proposed system?

53 responses

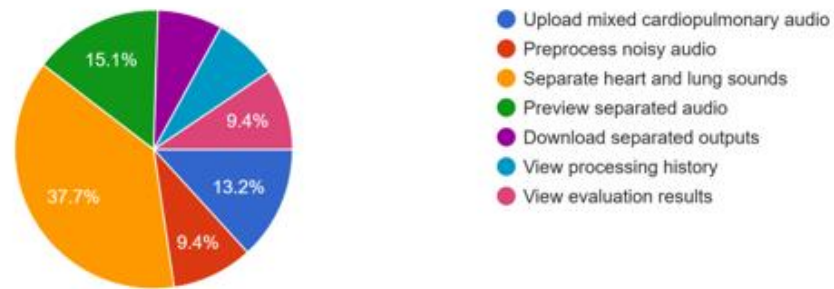


Figure 3.4: Preferred system features

Finding: The largest percentage of respondents selected separating cardiopulmonary recordings as the most desired system feature (20 or 37.7%). In second and third place were preview separated audio (8 or 15.1%) and uploading a mixed cardiopulmonary audio (7 or 13.2%). The other features received equal percentages of responses (5 or 9.4%) from the respondents. The view processing history feature received 4 or 7.5% of the responses. Similarly, a related history-page question asked the respondents to indicate whether the prototype should allow users to view previous uploads or processing details. The finding indicates a need for a history feature because 44 or 83.0% of the respondents selected yes.

Requirement implication: The cardiopulmonary separation prototype needs to allow users to upload a mixed cardiopulmonary audio, separate the audio into heart and lung sounds, preview the separated audio, and download the separated audio recordings. Furthermore, it would be desirable to allow users to view processing history if the prototype's database can allow tracking of the previously processed cardiopulmonary audio recordings.

3.2.7 Question 5: Preferred Output after Sound Separation

The fifth research question asked the respondents to identify their preferred output after cardiopulmonary separation. The options included receiving only the audio files, waveform visualisation, spectrogram visualisation, evaluation results, or a combination of any of the available outputs.

2. Which output would you prefer after sound separation?

53 responses

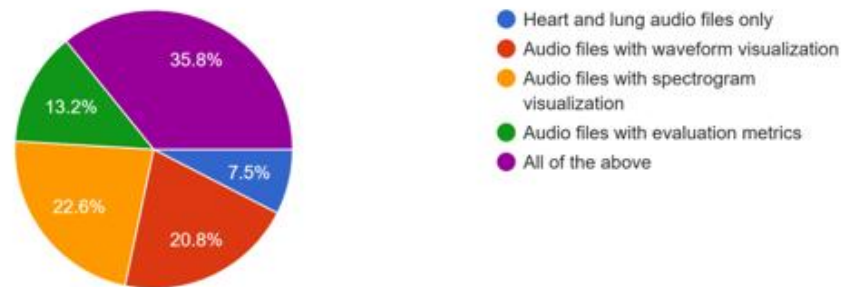


Figure 3.5: Preferred output after sound separation

Finding: 19 or 35.8% of the respondents selected all the available outputs (audio files, waveform, spectrogram, and evaluation results). Furthermore, 12 or 22.6% and 11 or 20.8% of the respondents selected the audio files with spectrogram and waveform options respectively. In addition, 7 or 13.2% of the respondents selected audio files with evaluation results, whereas 4 or 7.5% selected only the heart and lung audio files. A related result-display question from the cardiopulmonary separation prototype design section showed that 21 or 39.6% of the respondents selected an audio player that displays waveform, spectrogram, and evaluation results.

Requirement implication: The cardiopulmonary separation prototype should not rely on providing only the download link as the output display option. Rather, the prototype should display the separated audio files with options to preview the files as waveforms and spectrograms. In addition, the display should contain the evaluation results.

3.2.8 Question 6: Importance of Audio Preview Function

The sixth research question asked the respondents to evaluate the importance of a preview function for the separated heart and lung sounds before downloading.

3. How important is it for the system to provide a preview of the separated heart and lung sounds?
53 responses

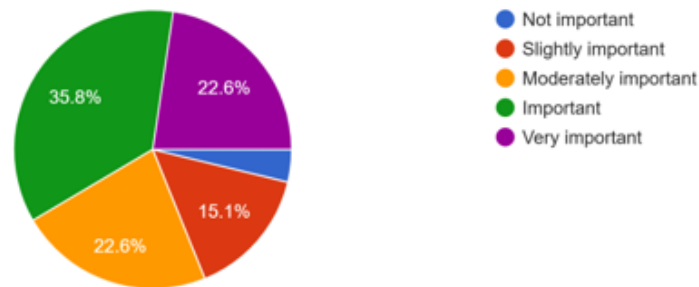


Figure 3.6: Preview separated heart and lung sounds

Finding: The largest group, 19 respondents (35.8%), rated audio preview as important. Another 12 (22.6%) selected very important and 12 (22.6%) selected moderately important. Eight respondents (15.1%) selected slightly important, while 2 (3.8%) selected not important.

Requirement implication: The cardiopulmonary separation prototype should allow users to preview the separated audio files as a preview function. A download option should also be available for the separated heart and lung audio files.

3.2.9 Question 7: Preferred Interface Style

The seventh research question asked the respondent to indicate their preferred interface style for a cardiopulmonary separation prototype. The options included a simple interface, a detailed interface, and a balanced interface.

2. Which interface style would you prefer?

53 responses

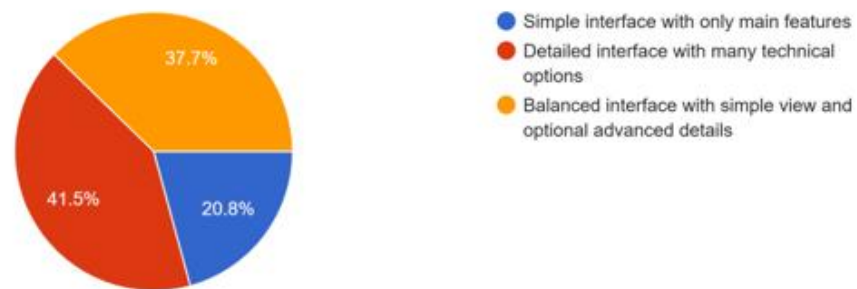


Figure 3.7: Preferred interface style

Finding: The largest group, 22 respondents (41.5%), preferred a detailed interface with many options. A balanced interface with a simple view and optional advanced settings was selected by 20 respondents (37.7%), while 11 (20.8%) preferred a simple interface with only basic options. In a related question, 33 respondents (62.3%) rated ease of use as moderately important, important or very important. For error feedback, 29 respondents (54.7%) preferred a message that explains the problem and suggests a correction.

Requirement implication: The prototype should provide a clear default workflow with advanced controls available when needed. Model selection, preprocessing information, evaluation results, waveforms and spectrograms can be shown as additional controls without crowding the main upload and result workflow.

3.2.10 Question 8: First Information Shown after Separation

The eighth research question asked the respondents to indicate the type of information that the cardiopulmonary separation prototype should display after the cardiopulmonary audio has been separated.

4. What information should appear first after separation is completed?

53 responses

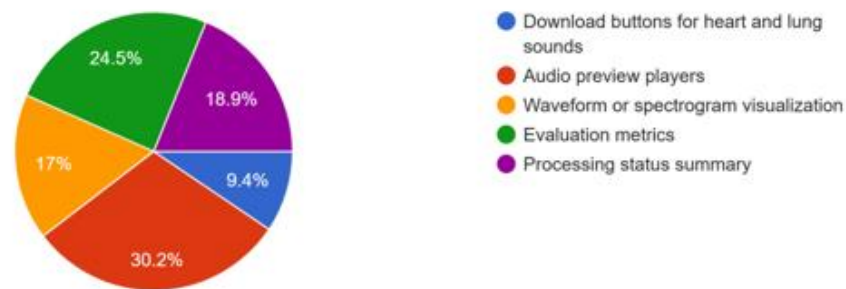


Figure 3.8: Preferred first information after separation is completed

Finding: Audio preview players were the preferred information to display after cardiopulmonary separation has been completed. This option received 16 or 30.2% of the responses from the respondents. Evaluation results received equal percentages (13 or 24.5%) of responses, whereas processing status summary received 10 or 18.9% of the responses. In addition, waveform or spectrogram received 9 or 17.0%, and download buttons received 5 or 9.4% of the responses. A related processing-information question in the cardiopulmonary separation prototype design section shows that 48 or 90.6% of the respondents rated file name, processing status, and processing time as moderately important, important, or very important.

Requirement implication: The cardiopulmonary separation prototype should display the separated cardiopulmonary audio files with preview players as the first information after the separation process is completed. The second type of information should be the evaluation results followed by the processing status summary, waveform/spectrogram display, and the availability of download buttons for the separated cardiopulmonary audio files.

3.2.11 Question 9: Importance of Evaluation Metrics

The ninth research question asked the respondents to rate the importance of displaying evaluation metrics for the separated cardiopulmonary audio files.

1. How important is it for the system to show evaluation metrics for the separated output?

53 responses

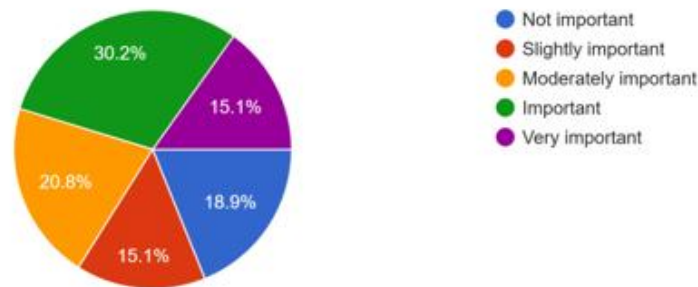


Figure 3.9: Importance of evaluation metrics for separated output

Finding: Sixteen respondents (30.2%) rated evaluation metrics as important, 11 (20.8%) selected moderately important and 8 (15.1%) selected very important. Eight respondents (15.1%) selected slightly important and 10 (18.9%) selected not important. A related question asked which information would be useful and listed signal quality, separation quality, noise reduction, processing time and an all-of-the-above option.

Requirement implication: The cardiopulmonary separation prototype should display evaluation information when valid evaluation results are available. For the FYP1, the evaluation requirements remain a design requirement because it has not been possible to implement a working cardiopulmonary separation model. As such, the cardiopulmonary separation prototype can display processing time, signal quality, separation quality, and noise reduction information to the extent that it is possible given the training data and evaluation dataset.

3.2.12 Question 10: Willingness to Use the Prototype

This question asks whether respondents would be willing to use a prototype that can separate mixed cardiopulmonary audio into heart and lung sound outputs. The related open-ended question is presented in text below the chart, instead, due to being improvement suggestions rather than specific feedback.

3. Would you use a prototype that separates mixed cardiopulmonary audio into heart and lung sound outputs?
53 responses

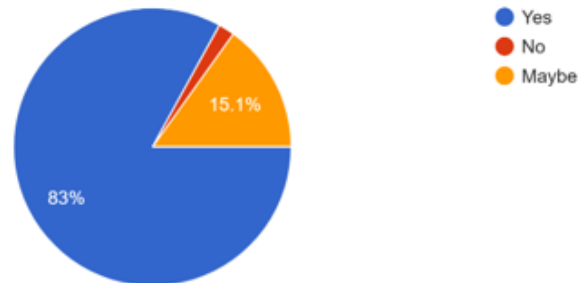


Figure 3.10: Willingness to use the cardiopulmonary sound separation prototype

Finding: 44 respondents (83.0% of answers) answered yes to the question if they would be willing to use a cardiopulmonary audio prototype able to separate the sounds of the heart and lungs. 8 respondents (15.1%) answered maybe, and 1 respondent (1.9%) answered no. The open-ended question had 18 non-empty responses. The suggestions were varied with some being vague or unrelated to the prototype, but the useful ones included requests for easier use by non-technical users, visualisation of the outputs as waveforms and spectrograms, preview and download features, evaluation metrics, more detailed explanations of results, improved noise reduction features, and better separation quality. Some responses mentioned diagnosis-related features, but they are out of the scope of this FYP, as the prototype only separates sounds without diagnostic capabilities.

Requirement implication: As shown by the high percentage of positive responses among respondents, there is a general willingness to use the described prototype. Therefore, the design should be further explored and developed, but later stages should include improvements suggested by the users. The scope of the prototype should be limited to cardiopulmonary sound separation with additional features such as improved separation quality, usability, visualisation, and evaluation being added in later iterations.

3.2.13 Target-User Validation Limitation and Follow-Up

The 53-response questionnaire remains useful for identifying general expectations such as simple upload, clear error messages, output preview,

download and evaluation information. It does not fully answer role-specific questions about who records the audio, which attributes must be entered, how files are handed to an analyst, or how previous recordings are retrieved in a healthcare setting.

Before the FYP2 requirements are frozen, a purposive follow-up should be conducted with healthcare personnel who handle or review body-sound recordings and with audio analysts or biomedical-signal researchers. The follow-up should confirm the minimum audio attributes, the audio-list workflow, role responsibilities, privacy expectations, result presentation and acceptable error messages. No responses from these target roles are invented in this report.

3.3 Functional Requirements

The initial set of the prototype's functional requirements is provided in the table below.

Table 3.3: Functional requirements for cardiopulmonary sound separation prototype

Req ID	Requirement	Description
F1	Enter Audio Attributes	The system shall allow healthcare staff to enter non-identifying attributes such as recording source, body location, device, sample rate, duration and notes.
F2	Upload Audio File	The system shall allow healthcare staff or an audio analyst to upload a mixed cardiopulmonary WAV file.
F3	Validate Audio and Attributes	The system shall validate the file type, header, readability, size and required attributes before processing.
F4	List Audio Files	The system shall display audio records with filename, attributes, upload time, selected model, status and result availability.
F5	Search or Filter Audio Records	The system shall allow the audio analyst to filter records by filename, status, model or supported attributes.
F6	Preprocess Audio	The system shall apply the preprocessing required by the selected separation strategy.
F7	Select Separation Model	The system shall allow the audio analyst to select an available separation model or use the active default.
F8	Run Sound Separation	The system shall execute the selected strategy on the validated and preprocessed recording.
F9	Generate Heart Sound Output	The system shall create a separated heart sound file.
F10	Generate Lung Sound Output	The system shall create a separated lung sound file.
F11	Display Processing Status	The system shall show validation, preprocessing, running, completed or failed status with a clear message.
F12	Preview Separated Outputs	The system shall allow authorised users to preview the heart and lung outputs.

Req ID	Requirement	Description
F13	Download Separated Outputs	The system shall allow authorised users to download the generated heart and lung files.
F14	View Processing History	The system shall retain a traceable list of audio records, jobs, selected models and output locations.
F15	Display Valid Evaluation Information	The system shall display evaluation metrics only when reference data or a valid evaluation procedure is available.

3.4 Non-Functional Requirements

The initial set of the prototype's non-functional requirements is provided in the table below.

Table 3.4: Non-functional requirements for cardiopulmonary sound separation prototype

Req ID	Category	Requirement
NF1	Usability	Healthcare staff shall be able to enter attributes and upload a file without configuring algorithm details.
NF2	Performance	The prototype shall show progress and avoid appearing unresponsive during processing.
NF3	Reliability	A failed job shall not be reported as completed, and incomplete outputs shall not be offered for download.
NF4	Maintainability	Validation, preprocessing, model execution, storage and result handling shall remain separate modules.
NF5	Security and Privacy	The prototype shall avoid unnecessary identifying information and restrict file access to the local application workflow.
NF6	Compatibility	The first prototype shall support WAV input and produce common downloadable audio output files.
NF7	Data Integrity	Audio attributes, selected model, job status and output paths shall remain linked to the correct record.
NF8	Scalability	The design shall allow more models and records to be added without changing the complete workflow.

3.5 User Requirements

The user requirements are summarized in the table below.

Table 3.5: User requirements for proposed prototype

User Requirement ID	User Need	Description
UR1	Healthcare staff enter recording details	Healthcare staff need to record non-identifying audio attributes before upload.
UR2	Healthcare staff upload a recording	Healthcare staff need a simple upload process with clear validity feedback.
UR3	Audio analyst reviews the audio list	The analyst needs a searchable list of records, attributes and processing status.
UR4	Audio analyst selects a model	The analyst needs to choose an available strategy or use the active default model.
UR5	Audio analyst starts separation	The analyst needs to run processing for a selected audio record and model.
UR6	Users understand job status	Both roles need clear progress, completion and error messages.
UR7	Users preview and download outputs	Both roles need to access separate heart and lung files after successful processing.
UR8	Audio analyst reviews history and metrics	The analyst needs traceable previous jobs and valid evaluation information when available.

3.6 Summary

This chapter used the 53 questionnaire responses as preliminary input and combined them with the literature, existing-application comparison, project scope and supervisor feedback. The requirements now define two concrete roles: healthcare staff and audio analyst. They also add audio attributes, an audio file list, filtering, role-specific model selection, traceable history and conditional evaluation information. Because the original questionnaire did not adequately represent hospital or clinic personnel, targeted validation with the intended users remains an FYP2 requirement rather than a completed result.

CHAPTER 4: SYSTEM DESIGN

4.1 Overview

This chapter defines an application-oriented design for the cardiopulmonary sound separation prototype. The design separates responsibilities between Healthcare Staff and Audio Analyst and covers audio attributes, upload, file listing, model selection, processing, output review, download and history. The diagrams and use cases describe a planned FYP2 solution rather than a completed hospital product, and no diagnostic function is included.

This design is inspired by one of my Software Design prototype projects in GitHub, which implements the minimal working set of a web interface, backend API, service modules, SQLite metadata storage, filesystem storage for audio data, and model selection. Similarly, the current report proposes analogous concepts for cardiopulmonary sound separation. Most importantly, this design is not expected to demonstrate actual cardiopulmonary sound separation: it defines the prototype to be built for FYP2 but not the separation results to be achieved at FYP2.

4.2 Context Diagram

The context diagram shows the proposed system and its external interactions. Healthcare staff enter non-identifying audio attributes, upload a mixed recording and access status or outputs. The audio analyst reviews audio records, selects a model, runs separation and inspects technical results. Approved public cardiopulmonary datasets support development and evaluation. The system boundary remains limited to heart-lung sound separation.

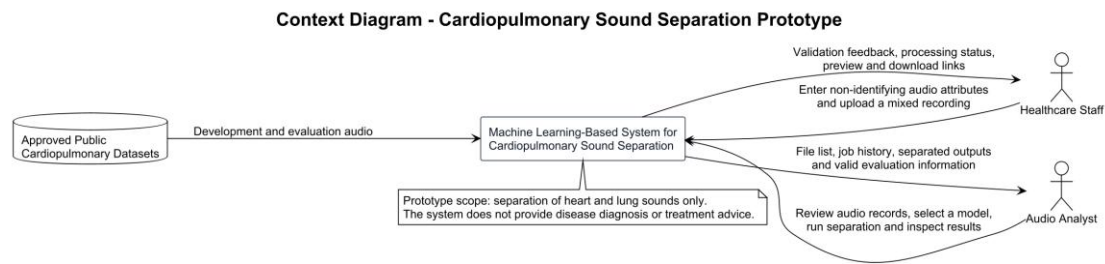


Figure 4.1: Context diagram for the proposed cardiopulmonary sound separation prototype

The cardiopulmonary sound separation prototype processes the user's mixed cardiopulmonary recording: it validates the input, selects the separation model, performs the separation, and stores/viewing the results. It is not supposed to help out with actual medical diagnosis or treatment, which is why the cardiopulmonary sound separation prototype does not implement disease detection or therapy recommendation features. The design is focused on the mixed cardiopulmonary audio specifics, where centre-based cardiopulmonary sound separation benefits from noise-aware processing of recordings with artefacts, overlapping sources, and diverse signal properties ([Grooby et al., 2023](#); [Fattahi et al., 2022](#); [Roy et al., 2024](#)).

4.3 Use Case Diagram

The use case diagram replaces the previous unspecified actor with two concrete roles. Healthcare Staff handles recording entry and upload, while Audio Analyst handles model selection, separation and technical review. Shared use cases cover status, preview and download. The diagram uses ten use cases that are described individually in the following tables.

Use Case Diagram - Cardiopulmonary Sound Separation Prototype

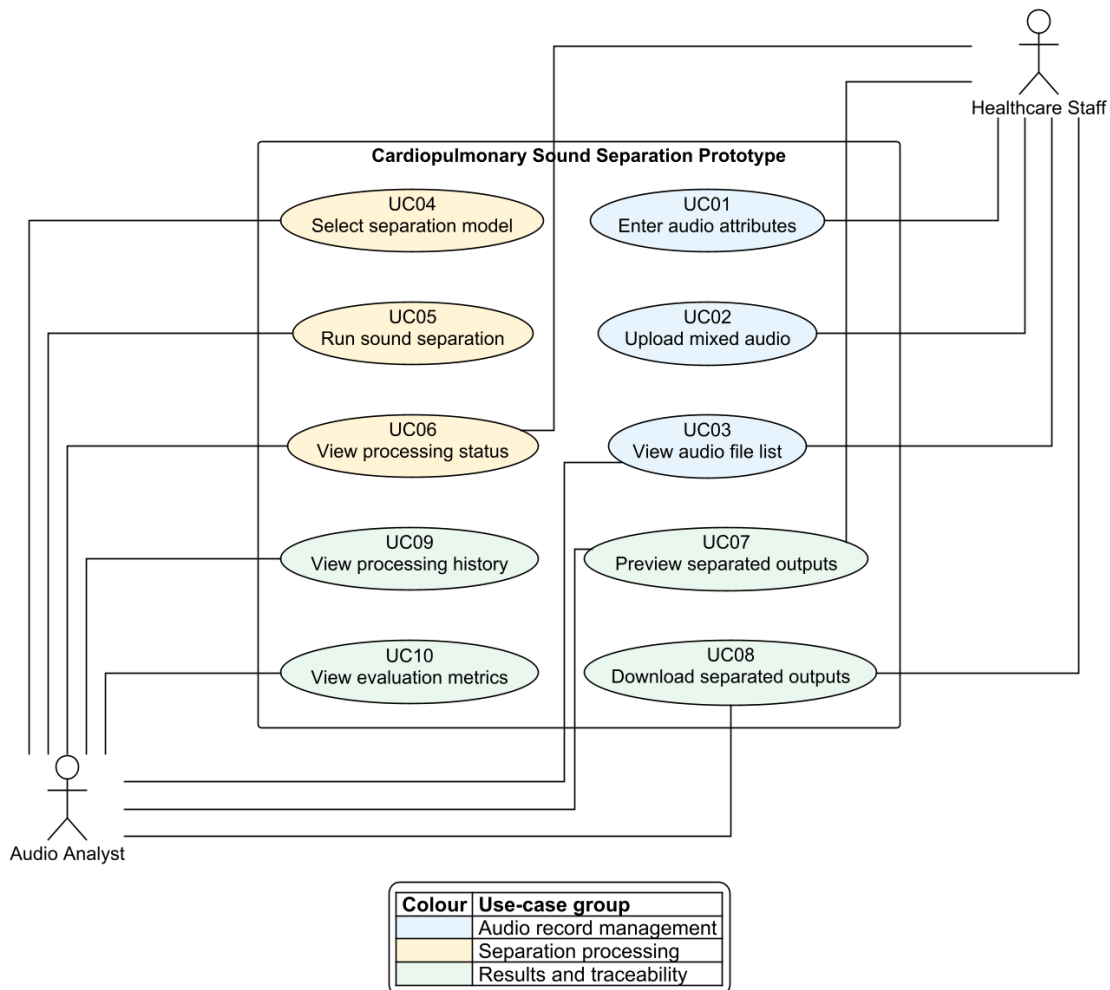


Figure 4.2: Use case diagram for the cardiopulmonary sound separation prototype

4.3.1 Use Case Description

4.3.1.1 Actors

Healthcare Staff is responsible for entering non-identifying recording attributes, uploading audio, following status and accessing completed outputs. Audio Analyst is responsible for reviewing the audio list, selecting a model, starting separation, checking technical results and reviewing history. The roles describe application responsibilities; they do not imply that the prototype performs diagnosis.

4.3.1.2 Use Case Description Tables

Each use case in the diagram is described separately below so that its actor, conditions, main flow and exception handling are explicit.

Table 4.1: Use case description for UC01 - Enter audio file attributes

Use case ID	UC01
Use case name	Enter audio file attributes
Primary actor	Healthcare Staff
Preconditions	The upload form is available and no patient-identifying information is required.
Trigger	Healthcare staff prepares a recording for upload.
Main flow	1. Open the upload form. 2. Enter supported attributes such as body location, recording device and notes. 3. Continue to audio selection.
Alternative / exception	Missing required attributes are highlighted before submission.
Postconditions	The attributes are ready to be saved with the uploaded audio record.

Table 4.2: Use case description for UC02 - Upload mixed cardiopulmonary audio

Use case ID	UC02
Use case name	Upload mixed cardiopulmonary audio
Primary actor	Healthcare Staff
Preconditions	A supported WAV file and the required non-identifying attributes are available.
Trigger	Healthcare staff submits the upload form.
Main flow	1. Select the WAV file. 2. Submit the form. 3. The system validates the file. 4. The system stores the file and creates an audio record.
Alternative / exception	An unsupported, unreadable or oversized file is rejected with a correction message.
Postconditions	A valid audio record appears in the audio file list with an uploaded status.

Table 4.3: Use case description for UC03 - View audio file list

Use case ID	UC03
Use case name	View audio file list
Primary actor	Audio Analyst; Healthcare Staff (limited view)
Preconditions	At least one audio record has been uploaded.
Trigger	The user opens the audio records page.
Main flow	1. Request the record list. 2. The system returns filenames, attributes, status and result availability. 3. The user filters or selects a record.
Alternative / exception	If no records exist, the system displays an empty-state message rather than an error.
Postconditions	A selected audio record is available for processing or review.

Table 4.4: Use case description for UC04 - Select separation model

Use case ID	UC04
Use case name	Select separation model
Primary actor	Audio Analyst
Preconditions	The selected audio record is valid and at least one model definition is active.
Trigger	The analyst opens the model selector for an audio record.
Main flow	1. Load active model definitions. 2. Display model name, version and strategy type. 3. Store the analyst's selection with the job request.
Alternative / exception	If no model is selected, the active default is used. If no active model exists, processing is blocked with an error.
Postconditions	A valid model is associated with the pending separation job.

Table 4.5: Use case description for UC05 - Run sound separation

Use case ID	UC05
Use case name	Run sound separation
Primary actor	Audio Analyst
Preconditions	A valid audio record and model selection are available.
Trigger	The analyst selects Run Separation.
Main flow	1. Create a job. 2. Preprocess the audio. 3. Resolve the configured strategy. 4. Run separation. 5. Save heart and lung outputs. 6. Mark the job completed.
Alternative / exception	Validation, preprocessing, model or output failure marks the job failed and records an explanatory message.
Postconditions	The job is completed with two output paths, or failed with traceable error information.

Table 4.6: Use case description for UC06 - View processing status

Use case ID	UC06
Use case name	View processing status
Primary actor	Healthcare Staff and Audio Analyst
Preconditions	A processing job exists.
Trigger	The user opens or refreshes the job page.
Main flow	1. Request the job state. 2. Display validation, preprocessing, running, completed or failed status. 3. Show the associated message and selected model.
Alternative / exception	If the job cannot be found, the system displays a record-not-found message.
Postconditions	The user understands the current job state and available next action.

Table 4.7: Use case description for UC07 - Preview separated outputs

Use case ID	UC07
Use case name	Preview separated outputs

Primary actor	Healthcare Staff and Audio Analyst
Preconditions	The job is completed and both output files exist.
Trigger	The user opens the completed result page.
Main flow	1. Load result metadata. 2. Display separate heart and lung audio players. 3. Allow the user to listen to each output.
Alternative / exception	A missing output disables its player and reports that the result is incomplete.
Postconditions	The user has reviewed the available outputs before download.

Table 4.8: Use case description for UC08 - Download separated outputs

Use case ID	UC08
Use case name	Download separated outputs
Primary actor	Healthcare Staff and Audio Analyst
Preconditions	The requested output exists for a completed job.
Trigger	The user selects Download Heart or Download Lung.
Main flow	1. Verify the job and output type. 2. Resolve the stored file path. 3. Return the selected audio file.
Alternative / exception	An unavailable output returns a clear message and no empty file is downloaded.
Postconditions	The selected heart or lung sound file is available on the user's device.

Table 4.9: Use case description for UC09 - View processing history

Use case ID	UC09
Use case name	View processing history
Primary actor	Audio Analyst
Preconditions	Previous audio records or jobs exist.
Trigger	The analyst opens the processing-history view.
Main flow	1. Request previous jobs. 2. Display audio attributes, model, status and completion time. 3. Open a selected result or failure record.
Alternative / exception	Filters that return no matches display an empty result set.
Postconditions	The analyst can trace previous processing activity without repeating the job.

Table 4.10: Use case description for UC10 - View evaluation metrics

Use case ID	UC10
Use case name	View evaluation metrics
Primary actor	Audio Analyst
Preconditions	The job is completed and valid reference data or an approved metric calculation is available.
Trigger	The analyst opens the evaluation area of a result.
Main flow	1. Retrieve stored metrics. 2. Display metric name, value and context. 3. Keep the audio outputs available for comparison.

Alternative / exception	If valid metrics are unavailable, the system states that no evaluation result is available and does not fabricate a score.
Postconditions	The analyst can review valid technical evaluation information without a diagnosis claim.

4.4 Activity Diagram

The activity diagram follows the role-based workflow from audio preparation to output access. Healthcare staff enter attributes and upload a recording. The system validates and stores the audio record. The audio analyst then selects the record and model, after which the system preprocesses the file and runs the chosen strategy. Completion requires both heart and lung output files; otherwise, the job is marked as failed with an explanatory message.

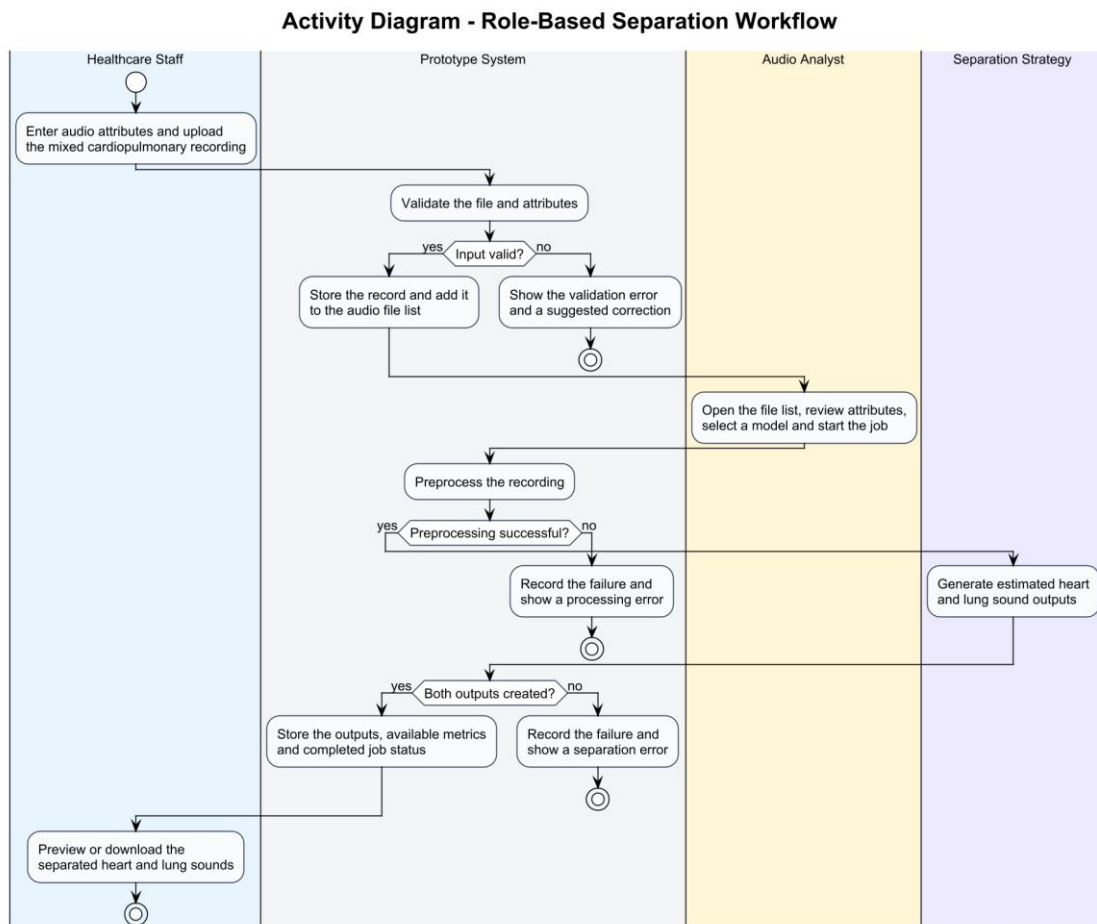


Figure 4.3: Activity diagram for the sound separation workflow

The activity diagram also specifies the system exit points, where the system is terminated gracefully (e.g., on an incompatible WAV file) or forcefully (e.g., on an unexpected error). Similarly, the system is able to recover from general file upload errors at the beginning of the workflow.

4.5 Class Diagram

Figure 4.4 below represents the class diagram for the cardiopulmonary sound separation prototype. It follows the service-oriented arrangement, where the interface interacts with route/controller classes, while the processing logic is encapsulated in service classes. In particular, the separation algorithm employs a strategy design pattern to separate mixed heart and lung sounds, which allows future modifications.

Class Diagram - Core Prototype Classes

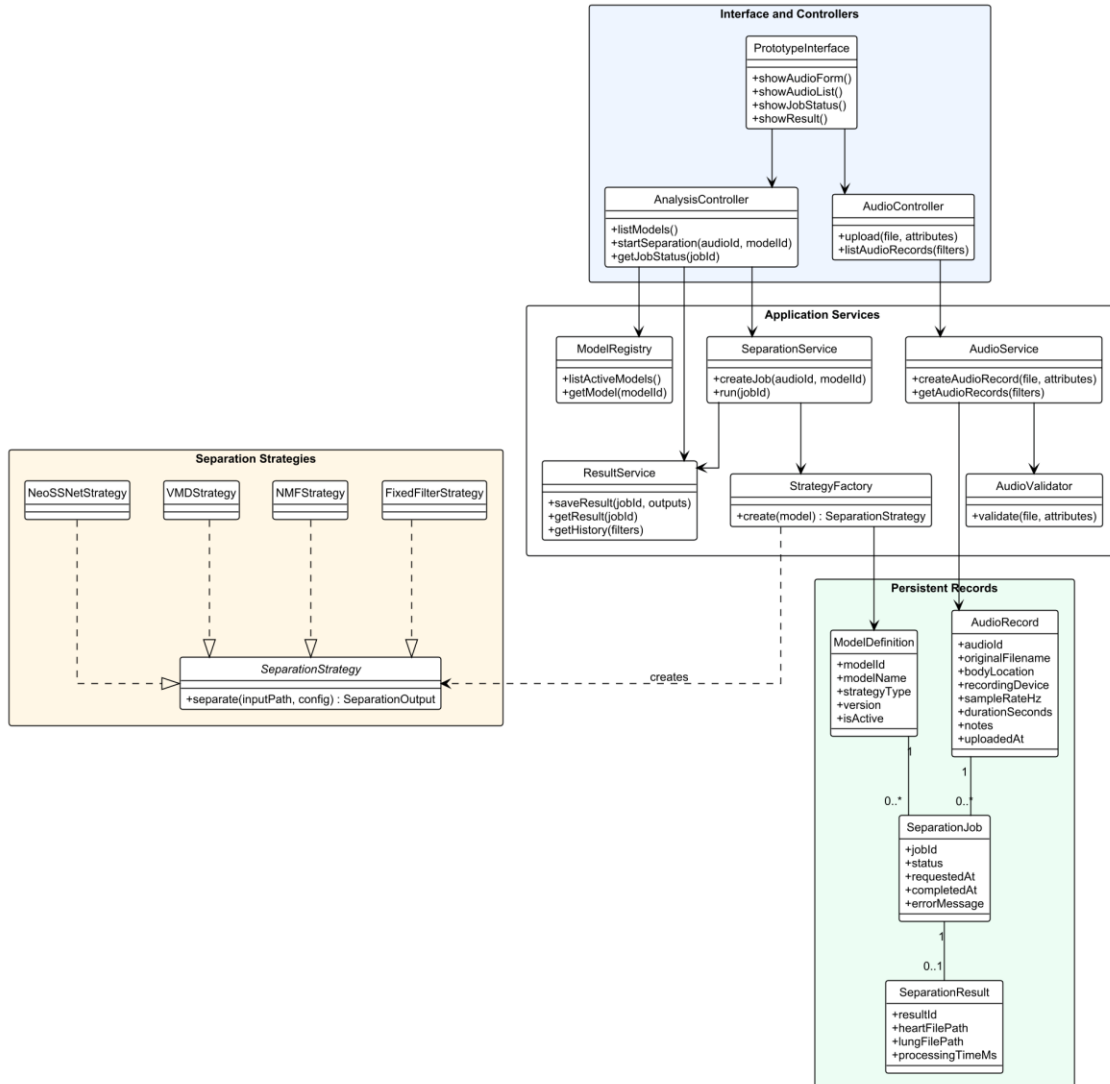


Figure 4.4: Class diagram for the cardiopulmonary sound separation prototype

4.5.1 Component-Level Architecture

Figure 4.5 below represents the component-level architecture for the cardiopulmonary sound separation prototype. It incorporates web and API interfaces on the client side, service and model components on the server side, and database and file storage components.

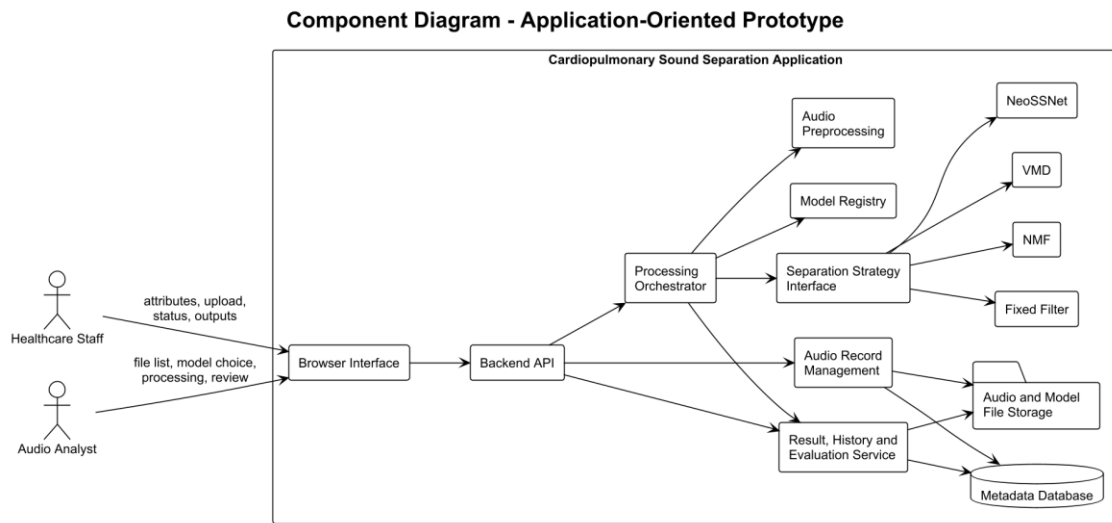


Figure 4.5: Component-level architecture for the cardiopulmonary sound separation prototype

Table 4.11 below provides an overview of the cardiopulmonary sound separation prototype components and their responsibilities.

Table 4.11: Main system modules and responsibilities

Module	Responsibility
User interface	Supplies upload controls, model selection, processing status, result display, preview controls, downloads, and history view
Upload and validation module	Validated WAV input, parsed headers, extracted metadata, and ensured file integrity
Model service	Managed model registry, tracked default model, and validated model checkpoint and configuration paths
Preprocessing module	Converted uploaded audio into a model-specific waveform via mono conversion, resampling, normalisation, segmentation, and time-frequency transformations
Separation service	Coordinated the end-to-end processing workflow from upload to result delivery
Algorithm factory	Abstracted the model-specific implementation details and managed strategy instantiation
Separation strategies	Implemented specific algorithm stratagems, including baseline filtering, NMF/VMD, or NeoSSNet-style deep learning approaches
Result service	Stored result metadata, exposed download locations, retrieved result details, and supplied history information
Database module	Stored model records, job details, result metadata, system logs, and evaluation metrics
Filesystem storage	Saved uploaded mixed audio, generated heart and lung recordings, model files, and system logs

4.5.2 Algorithm and Model Selection Design

Each strategy is expected to follow the same interface to supply flexibility in selecting the algorithm for heart and lung separation. In particular, the strategy is instantiated with a mixed waveform as input and is expected to return two files or two waveforms as outputs: one corresponds to the heart sound, and the other corresponds to the lung sound. This approach is flexible because it would suffice to implement a single strategy for the initial cardiopulmonary sound separation prototype iteration while leaving the option to expand the model selection in later FYP2 iterations without reworking the interface, result storage, and history display logic.

The algorithm strategies are informed by the literature review on analysis of cardiopulmonary recordings, decomposition methods for mixed signal processing, unsupervised separation methods for heart and lung sounds, and the NeoSSNet-style deep learning paradigm ([Grooby et al., 2023](#); [Fattahi et al., 2022](#); [Sun et al., 2024](#); [Ullah & Zhang, 2024](#); [Poh et al., 2024](#)).

Table 4.12 below lists the strategies to be considered for model selection in FYP2. Each strategy is further described below the table to explain the anticipated contribution to the cardiopulmonary sound separation prototype. Note that the actual results are not described because they will be available in FYP2.

Table 4.12: Options for model selection in the cardiopulmonary sound separation prototype

Strategy	Role in Prototype	Reason for Inclusion
Fixed filter baseline	Simple baseline to verify that the prototype can process, store, and display separated signals	Limited utility due to overlapping heart and lung responses, but it represents a viable option when these responses are non-overlapping
NMF separation strategy	Fundamental matrix-factorisation baseline or fallback option	Enables comparison with deep learning approaches and provides interpretability
VMD separation strategy	Fundamental mode-decomposition baseline or fallback option	Supplies another decompositional approach for analysing mixed cardiopulmonary recordings
NeoSSNet deep separation strategy	Primary machine learning strategy for heart-lung separation	Aligns with the project title and focuses on the selected direction of research

Future model versions	Optional extension in FYP2 beyond the initial strategy	Allows additional model checkpoints, strategies, or algorithms to be added without complicating the prototype architecture
-----------------------	--	--

The model registry will be implemented as a model selection UI, where each registry entry will store the model name, version, architecture, framework, checkpoint path, configuration path, description, and status. Upon model selection, this information will be stored in the separation job to track the model version. If no model is selected, the system will utilise the default separation strategy.

4.5.3 Separation Algorithm Design and Mathematical Formulation

The algorithm design is placed in Chapter 4 because it defines how the proposed solution will transform one mixed recording into estimated heart and lung outputs. The strategies share one application workflow and one output contract, but they differ in how they represent and separate the signal. The equations and strategy descriptions below are design specifications for FYP2, not completed performance results.

In particular, the mixed cardiopulmonary signal $x[n]$ can be expressed as:

$$x[n] = h[n] + l[n] + r[n],$$

where $h[n]$, $l[n]$, and $r[n]$ are the heart, lung, and residual components, respectively. The goal of the separation strategy is to estimate $h[n]$ and $l[n]$ given the mixed $x[n]$.

4.5.3.1 Input Preparation

The input preparation step is essential to maintain uniformity: the recorded cardiopulmonary sounds can contain a variety of noise, artefacts, and preparation issues, which will undermine the separation process ([Grooby et al., 2023](#); [Fattahi et al., 2022](#)).

A generic input preparation workflow could look like this:

1. Read the uploaded file.
2. Convert stereo files to mono if necessary.
3. Resample to the target sample rate.
4. Perform amplitude normalization.

5. Apply segmentation or transformation if necessary.

Amplitude normalization could be performed according to the following formula:

$$\tilde{x}[n] = \frac{x[n]}{\max_n |x[n]| + \epsilon},$$

where $\text{amp_scale} = \text{max_amp} / (\text{max_amp} - \text{min_amp} + \text{EPS})$ with EPS being a small value to avoid division by zero. Short-time Fourier transform (STFT) can be applied if the separation strategy requires a time-frequency representation:

$$X(k, m) = \sum_{n=0}^{N-1} \tilde{x}[n] w[n - mH] e^{-j2\pi kn/N},$$

where $w[n]$ is the window function, H is the hop size, N is the FFT size, k is the frequency bin, and m is the frame index.

4.5.3.2 Fixed Filter Baseline

The fixed filter baseline is a simple approach that can be used as a comparison baseline and a preparatory test for more involved decompositions ([Roy et al., 2024](#); [Zhao et al., 2024](#)). It implements rough frequency filtering with fixed masks or filters to isolate the heart and lung sounds:

$$\hat{H}(k, m) = M_h(k) X(k, m),$$

$$\hat{L}(k, m) = M_l(k) X(k, m),$$

with $(M_h(k))$ and $(M_l(k))$ being the fixed masks or filters for the heart-dominated and lung-dominated frequencies, respectively. The time-domain estimates are reconstructed using the inverse STFT:

$$\hat{h}[n] = \text{iSTFT}(\hat{H}(k, m)), \quad \hat{l}[n] = \text{iSTFT}(\hat{L}(k, m)).$$

This algorithm is not expected to perform well on strongly overlapping cases, but the actual utility of such a trivial algorithm is to sanity check that the upload, preprocessing, output writing, preview, and download procedures work before proceeding to a more involved model.

4.5.3.3 NMF and VMD-Based Decomposition

Non-negative matrix factorization (NMF) and variational mode decomposition (VMD) can be suggested as two particular traditional approaches. The decomposition methods are suitable for providing explainable baselines for cardiac-lung separation and comparison ([Sun et al., 2024](#); [Ullah & Zhang, 2024](#)). They could be implemented in the proposed design as NmfSeparationStrategy and VmdSeparationStrategy classes or particular cases of the same interface.

For the NMF strategy, if ($V = |X|$) is the non-negative magnitude spectrogram, NMF approximates it as:

$$V \approx WH,$$

where (W) contains spectral basis patterns and (H) contains their time activations. A common reconstruction loss takes the form of:

$$\min_{W, H \geq 0} D(V || WH) + \lambda R(W, H),$$

where $D(\cdot)$ is a divergence measure and $R(\cdot)$ is a regularisation term. Separate heart- and lung-specific estimates can be obtained by defining soft masks:

$$M_h = \frac{V_h}{V_h + V_l + \epsilon}, \quad M_l = \frac{V_l}{V_h + V_l + \epsilon}.$$

and separate the contents of the spectrogram by:

$$\hat{H} = M_h \odot X, \quad \hat{L} = M_l \odot X,$$

Decomposition can then be applied to the time series directly to isolate the heart and lung sounds. Alternatively, VMD can be applied to the signal $x[n]$ to obtain the modes $u_k(t)$ with centre frequencies f_k :

$$\min_{\{u_k\}, \{\omega_k\}} \sum_k \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2$$

such that:

$$\sum_k u_k(t) = x(t).$$

The modes can then be separated into the heart- and lung-specific components by grouping the ones with similar centre frequencies.

4.5.3.4 NeoSSNet Deep Separation Strategy

NeoSSNet-style deep separation strategy will be the core machine learning model, as it was designed for the cardiopulmonary separation and can be employed for this task directly ([Poh et al., 2024](#)). It is expected to utilise the input from the preparatory steps and estimate the two output channels with a neural network:

$$[\hat{h}, \hat{l}] = f_{\theta}(\tilde{x}),$$

where f_{θ} represents the model with parameters θ . If the separation strategy involves time-frequency masks, they can be represented as:

$$[M_h, M_l] = g_{\theta}(X),$$

$$\hat{H} = M_h \odot X, \quad \hat{L} = M_l \odot X.$$

If the references are available for the training data or the evaluation set, a reconstruction loss can be defined as:

$$\mathcal{L}_{rec} = \|h - \hat{h}\|_1 + \|l - \hat{l}\|_1.$$

For cardiopulmonary separation, the SI-SDR between the estimate y and the reference source s can also be used if the relevant information is available ([Torabi et al., 2025a](#); [Zhao et al., 2024](#)). For one estimated source ($\hat{s}$) and reference source (s):

$$s_{target} = \frac{\langle \hat{s}, s \rangle}{\|s\|^2} s,$$

$$e_{noise} = \hat{s} - s_{target},$$

$$SI - SDR = 10 \log_{10} \frac{\|s_{target}\|^2}{\|e_{noise}\|^2}.$$

The first FYP2 target is working inference path from upload to separated output files. After that, the training, fine-tuning, or model replacement can be attempted if enough data, suitable hardware, and supervisor's approval are available.

4.5.3.5 Model Selection Strategy

A model choice strategy can be implemented as a strategy to select the algorithm to use on upload time: the user will be able to choose the model from a list. The controller will pick the model-specific strategy if present in the request or use the default one. Factory methods will be used to tie the model configuration to the strategy:

```
strategy = Factory(model.architecture).
```

For example, for a specific architecture NeoSSNet, you can make a model record, which would create a NeoSSNet strategy, and specific names FixedFilter, NMF, VMD would create their own concrete strategies. By this approach, you can easily extend the system by adding new algorithms and strategies - it will require only creating a new strategy class and entry in the model registry, instead of changing the whole application.

4.5.4 Database and Storage Design

The database design uses relational tables for metadata and file-system storage for binary audio. SQLite is suitable for the first local prototype because it is lightweight and easy to inspect during iterative development. Audio records, model definitions, processing jobs and separation results remain linked so that uploads and outputs can be traced.

Table 4.13 below provides an overview of the database entities that will be used for the cardiopulmonary sound separation prototype.

Table 4.13: Database entities for prototype traceability and analysis

Entity	Description
uploaded_audio	Table to store metadata about each uploaded mixed WAV file, including filename, storage path, sample rate, channels, and duration
model	Table to store information about available models or algorithms, including their names, architectures, checkpoint paths, configuration paths, and statuses
separation_job	Table to track details about each separation request, including the model, status, timestamps, processing duration, parameters, and error messages
separation_result	Table to store metadata about processed heart and lung sound recordings
evaluation_metric	Table to store optional evaluation metrics if reference signals are available for ground truth assessment

system_log	Table to store system-level logs about uploads, processing, results retrieval, and other events
------------	---

The file system will be organised into subdirectories to store uploaded files, temporary files, heart sound recordings, lung sound recordings, model files, and system logs. This organisation will supply flexibility for data inspection and cleanup procedures because it will be relatively straightforward to delete demonstration data and differentiate public test data from prototype files.

4.6 Sequence Diagram

The sequence diagram focuses on the Audio Analyst's model-selection and separation use case. Every lifeline corresponds to a class or interface defined in the class diagram. `PrototypeInterface` passes the selected audio record and model to `AnalysisController`. `ModelRegistry` supplies the model definition, `SeparationService` creates and runs the job, `StrategyFactory` resolves a `SeparationStrategy`, and `ResultService` stores the completed result for preview, download or later review.

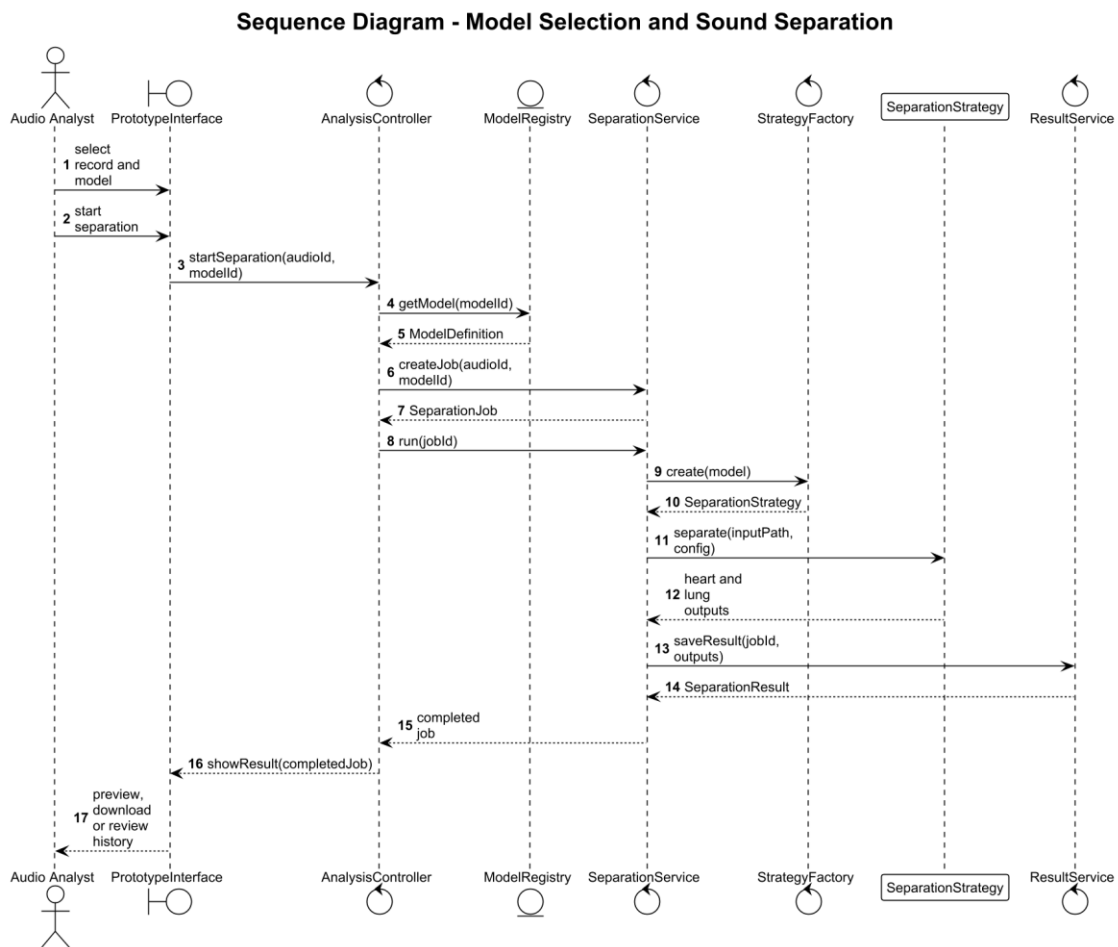


Figure 4.6: Sequence diagram for model selection and cardiopulmonary sound separation

The sequence diagram is simplistic to reflect a minimal prototype design that can be implemented in FYP2. In particular, it is possible to execute this cardiopulmonary sound separation prototype in a synchronous manner for a short demonstration without a dedicated job queue. By contrast, the class- and component-level designs leave the option open for further production optimisations, including asynchronous job execution.

4.7 Interface Design

The interface is organised around the two application roles. The upload view provides non-identifying audio attribute fields, a WAV selector and validation feedback for healthcare staff. The audio analyst view provides an audio file list, filters, model selection, processing status, result preview, download controls,

history and valid evaluation information. The wording must consistently describe sound separation rather than diagnosis.

Table 4.14 below summarises the interface areas for cardiopulmonary sound separation.

Table 4.14: Interface areas for cardiopulmonary sound separation

Interface Area	Description
Upload area	WAV file selector, note about the accepted format, selected filename, and Start button
Model selection area	Dropdown list with available separation strategies, default model indicator, and strategy description
Status area	API status, upload validation status, preprocessing status, model execution status, completion time, and error messages
Result area	Preview controls for heart and lung sounds, result file download controls, and optional evaluation display
History area	List of recent jobs with filename, status, duration, and options to reopen or download results

No interface mockups are produced for this FYP1 report because interface design and user testing belong to FYP2. However, the interface areas inform the questionnaire in Chapter 3, particularly the result preview options, file download instructions, status displays, and evaluation viewport.

4.8 Summary

The revised system design treats the project as a role-based application. It specifies how healthcare staff and audio analysts work with audio records, how each use case behaves, how data and results remain traceable, and how multiple separation strategies fit behind one interface. The detailed algorithm design is now contained in this chapter. Chapter 5 therefore focuses only on the implementation, testing, deployment and documentation activities planned for FYP2.

CHAPTER 5: IMPLEMENTATION PLAN

5.1 Development Phase

FYP2 will implement the application design defined in Chapter 4. Work will begin with the role-based audio-record workflow, including attributes, upload, validation and the audio file list. The backend, storage and model registry will then be connected to a separation strategy. Result preview, download, history and conditional evaluation information will be added after the core path is stable. This chapter describes planned activities only and does not claim completed model or user-test results.

5.1.1 Front-End Development

The front end will provide separate task paths for healthcare staff and audio analysts within one prototype interface. Healthcare staff need a simple audio attribute and upload form. Audio analysts need a record list, filters, model selection, job controls and result review. Shared views will provide status, preview and download functions. The interface will avoid diagnosis wording because the prototype separates sounds only.

Instead, the following areas are going to be implemented:

1. The upload area: non-identifying audio attribute fields, a WAV selector and validation feedback
2. The audio file list: searchable records with attributes, status, selected model and result availability
3. The analyst controls: model selection, process initiation, status and clear failure messages
4. The result area: separate heart and lung previews, download buttons and valid metric information
5. The history area: previous jobs that can be filtered and reopened for review

The interface should make the separation-only scope clear and provide understandable status information. It should also display useful messages for unsupported files, unavailable models, preprocessing failures and separation failures.

5.1.2 Back-End Development

The back end shall process the uploaded files, validate the audio, choose and execute the separation algorithm, store and retrieve the results, and inform the front end about the status of the process. The back end is planned to be a thin controller with route-specific services (see Table 5.1). The routing logic is responsible for processing the request and returning an HTTP response, and the services define the business logic of file storage, model selection, job lifecycle, and result retrieval.

The table below outlines the planned API endpoints and their purpose.

Table 5.1: Planned backend API responsibilities

Endpoint or function	Purpose
GET /health	Check if API and database are online
GET /models	Get list of available models and default configuration
POST /upload	Upload file and validation
POST /separate/{audio_id}	Execute separation
GET /result/{job_id}	Get job status, metadata, links to results and timing
GET /download/{job_id}/heart	Download separated heart
GET /download/{job_id}/lung	Download separated lung
GET /history	Get history of recent jobs for history screen in UI

The back end is supposed to decouple model execution and routing, so that FYP2 would be able to implement the alternative strategy and update the model execution independently from the upload and result APIs.

5.1.3 Machine Learning Model Integration

The separation algorithms and their mathematical design are defined in Chapter 4. The FYP2 activity is to implement the shared strategy interface, connect the selected baseline and machine learning strategies, and verify that each configured strategy returns separate heart and lung output files through the same application workflow.

Implementation will begin with an end-to-end path using a baseline strategy. NMF, VMD and a NeoSSNet-style strategy can then be integrated according to data, hardware and supervisor approval. Training, fine-tuning and final metric reporting will be completed only when valid datasets and reference signals are available.

5.1.4 Database and Storage Integration

The prototype will use a lightweight database for metadata and file-system storage for audio. The database should store upload attributes, model metadata, selected model ID, job status, output paths, processing time, optional evaluation metrics and system logs. The file system will store uploaded recordings, model files and generated outputs.

The storage folders are:

```
storage/uploads/raw/  
storage/uploads/temp/  
storage/outputs/heart/  
storage/outputs/lung/  
storage/ml_models/  
storage/logs/
```

This design is practical and a good choice for a local academic prototype because it minimizes the database's size footprint while still allowing traceability of uploaded files, selected model records, processing jobs, and resulting outputs.

5.2 Testing

The test phase will focus on ensuring that the prototype works as designed and that the individual strategies are safe to use within the application. The test activities are described below, but they are planned rather than executed: the actual test execution will happen in FYP2.

5.2.1 Unit Testing

The unit tests will be written for generic modules and individual strategies. In particular, WAV validation, metadata extraction, path management, model lookup, default model selection, and strategy factory will be added. The separation strategy unit tests should not be checking for specific separation quality. Instead, they should make sure that the strategy correctly initiates and terminates in case of an incorrectly prepared input. The tests should also ensure that the strategy returns correct file paths for heart and lungs sounds and does not falsely report a finished job.

5.2.2 Integration Testing

The integration tests will ensure that the front end, API, database, storage, preprocessing, model selection, strategy factory, separation process, and result retrieval are compatible and work together. The following cases will be covered:

1. A user uploads a WAV file, and the metadata is stored in the database.
2. The model list is requested, and the default model is selected.
3. A separation process is initiated with a specific model ID.
4. The job records the selected model.
5. The successful job has heart and lung output paths recorded in the database.
6. The result and download endpoints return the correct job information.
7. The failed job with missing model files or an unsupported model architecture returns an informative error message.

5.2.3 System Testing

The system tests will follow the user interaction scenarios and attempt to repeat them to ensure that the prototype behaves as expected. In particular, the tests will include valid upload, unsupported file upload, invalid WAV header, model list retrieval, model-specific job execution, default model fallback, result preview, output download, history retrieval, and failure recovery. The tests will also check the user interface elements: the progress notifications, failed job display, unavailable download buttons, and the absence of medical diagnosis language.

5.2.4 User Acceptance Testing

User acceptance testing will be conducted after the prototype workflow is stable. Healthcare staff will be asked to complete audio-attribute entry, upload, status and output-access tasks. Audio analysts will be asked to find a record, select a model, run separation, review results and reopen a previous job. These activities are planned for FYP2 and have not yet been completed.

5.3 Deployment

The deployment phase will prepare the prototype for presentation and evaluation. The initial deployment target is the local machine because it will entail fewer security and hosting considerations: the tool will only process the uploaded cardiopulmonary recordings, separate them with the selected model, and store the results on the file system. The deployment instructions will describe how to set up the environment, launch the prototype, process an example WAV file, and review the results. In particular, the following steps are planned:

1. Setting up the Python environment.
2. Installing the backend and all the necessary machine learning libraries.
3. Preparing a database.
4. Making sure the storage location is available.
5. Putting the weights and configuration files in the working directory.
6. Starting the backend and the browser interface.
7. Uploading a short WAV file as a test.
8. Checking the progress and making sure that heart and lung files have been created.

If the prototype is later deployed elsewhere, additional work will have to be performed to host the API, secure the file storage, limit access to sensitive parts, manage long-running jobs, and monitor performance. These topics are outside the scope of the initial FYP1 effort, but they can be noted as future considerations.

5.4 Documentation and Maintenance Plan

I will document the deployment process, provide notes for end-users, describe how the API should be used, cover test coverage notes in the documentation, and maintain a distinction in the report between what was implemented, what was intended but not implemented, and what remains planned. The maintenance will involve updating the libraries and dependencies, providing more error context, implementing separation strategies, improving model selection, updating the model weights, improving preprocessing, implementing waveform display, evaluating the separation, and performing other toolbox-specific maintenance tasks.

5.5 Summary

Chapter 5 now describes the FYP2 activities at implementation-plan level. The work covers the role-based front end, backend services, audio attributes, file listing, database and storage integration, strategy integration, planned testing, deployment and documentation. Technical algorithm details have been moved to Chapter 4. No separation quality, testing outcome or healthcare acceptance result is reported as completed.

CHAPTER 6: CONCLUSION

6.1 Overview

This chapter concludes the FYP1 report by summarising the background study, literature review, requirements analysis, system design and implementation plan prepared for the cardiopulmonary sound separation prototype.

6.2 Project Summary

The FYP1 report establishes the basis for a role-based cardiopulmonary sound separation application. It connects the literature and questionnaire findings to a workflow in which healthcare staff prepare and upload audio records while an audio analyst selects a model, runs separation and reviews traceable outputs. The design remains a prototype specification for FYP2 rather than a completed clinical system.

The planned FYP2 workflow covers audio upload, validation, preprocessing, model selection, separation, preview, download and processing history. The FYP1 requirements and design define how these functions should work together before implementation begins.

6.3 Expected Contributions

The expected contribution is a reusable application workflow for managing and separating mixed cardiopulmonary recordings. In addition to producing heart and lung output files, the prototype is expected to manage audio attributes, validation, file lists, model selection, status, history and result access. The strategy interface also allows fixed-filter, NMF, VMD and NeoSSNet-style approaches to be compared without redesigning the surrounding application.

At the FYP1 stage, the completed contribution consists of the background study, requirements analysis, design artefacts and implementation plan needed to guide development and evaluation during FYP2.

6.4 Limitations

The current work is limited to FYP1 design and planning. The 53 questionnaire responses mainly represent academic and technical respondents, so they do not validate a hospital workflow. Model implementation, separation performance, healthcare-user validation, security testing and deployment remain incomplete. The prototype is also limited to sound separation and must not be presented as a diagnostic system.

The prototype is limited to separating mixed cardiopulmonary recordings into heart and lung sound outputs. It will not provide a diagnosis, treatment recommendation or clinical decision.

6.5 Future Work

FYP2 should first confirm the revised requirements with healthcare staff and audio analysts. Development can then implement audio attributes, the audio file list, model selection and the complete output workflow before improving the separation strategies. Later work may add waveform or spectrogram displays, annotations and stronger access control after the core application is working and has been tested honestly.

6.6 Summary

The FYP1 work defines an application-based prototype with two user roles, a traceable audio-record workflow, selectable separation strategies and a practical FYP2 implementation plan. Implementation, target-user validation, testing and performance evaluation remain future work and are not claimed as completed results.

References

- 3M. (n.d.). *3M Littmann CORE digital stethoscope*. Retrieved July 31, 2026, from <https://www.littmann.com/en-us/home/f/b5005222000/>
- Ali, S. N., Shuvo, S. B., Al-Manzo, M. I. S., Hasan, A., & Hasan, T. (2023). An end-to-end deep learning framework for real-time denoising of heart sounds for cardiac disease detection in unseen noise. *IEEE Access*, 11, 87887–87901. <https://doi.org/10.1109/access.2023.3292551>
- Ali, S. N., Zahin, A., Shuvo, S. B., Nizam, N. B., Nuhash, S. I. S. K., Razin, S. S., Sani, S. M. S., Rahman, F., Nizam, N. B., Azam, F. B., Hossen, R., Ohab, S., Noor, N., & Hasan, T. (2026). BUET multi-disease heart sound dataset: A comprehensive auscultation dataset for developing computer-aided diagnostic systems. *Computer Methods and Programs in Biomedicine Update*, 9, Article 100237. <https://doi.org/10.1016/j.cmpbup.2026.100237>
- Arjoune, Y., Nguyen, T. N., Salvador, T., Telluri, A., Schroeder, J. C., Geggel, R. L., May, J. W., Pillai, D. K., Teach, S. J., Patel, S. J., Doroshov, R. W., & Shekhar, R. (2023). StethAid: A digital auscultation platform for pediatrics. *Sensors*, 23(12), Article 5750. <https://doi.org/10.3390/s23125750>
- Baeyens, R., Ragusa, D., Stas, T., Ides, K., Marenzi, E., Leporati, F., Daems, W., & Steckel, J. (2025). Compressed sensing of acoustic cardiopulmonary signals using a CNN-based reconstruction method. *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 1–7. <https://doi.org/10.1109/embc58623.2025.11253482>
- Behera, S., Jain, P. K., & Bakshi, S. (2026). Adaptive thresholding of DWT coefficients using UNet for denoising real-life respiratory sounds. *Biomedical Signal*

Processing and Control, 111, Article 108232.

<https://doi.org/10.1016/j.bspc.2025.108232>

Eko Health. (2025). *Overview of the Eko app*. <https://support.ekohealth.com/hc/en-us/articles/10303967065883-Overview-of-the-Eko-App>

Fattahi, D., Sameni, R., Grooby, E., Tan, K., Zhou, L., King, A., Ramanathan, A., Malhotra, A., & Marzbanrad, F. (2022). A blind filtering framework for noisy neonatal chest sounds. *IEEE Access*, 10, 50715–50727.

<https://doi.org/10.1109/access.2022.3170052>

Grooby, E., Sitaula, C., Fattahi, D., Sameni, R., Tan, K., Zhou, L., King, A., Ramanathan, A., Malhotra, A., Dumont, G. A., & Marzbanrad, F. (2022). Real-time multi-level neonatal heart and lung sound quality assessment for telehealth applications. *IEEE Access*, 10, 10934–10948.

<https://doi.org/10.1109/access.2022.3144355>

Grooby, E., Sitaula, C., Fattahi, D., Sameni, R., Tan, K., Zhou, L., King, A., Ramanathan, A., Malhotra, A., Dumont, G., & Marzbanrad, F. (2023). Noisy neonatal chest sound separation for high-quality heart and lung sounds. *IEEE Journal of Biomedical and Health Informatics*, 27(6), 2635–2646.

<https://doi.org/10.1109/jbhi.2022.3215995>

Han, B., & Quan, W. (2025). Cardiorespiratory sound separation using singular spectrum analysis. *2025 17th International Conference on Signal Processing Systems (ICSPS)*, 731–735. <https://doi.org/10.1109/icspss66615.2025.11347745>

Hoang, V. T., Nguyen, Q., Nguyen, Q. C., Nguyen, X. P., & Nguyen, H. (2023). Sound-Dr: Reliable sound dataset and baseline artificial intelligence system for respiratory illnesses. *PHM Society Asia-Pacific Conference*, 4(1).

<https://doi.org/10.36001/phmap.2023.v4i1.3604>

- Jakubec, M., Lieskovska, E., & Pocta, P. (2025). A robust deep learning based model for denoising phonocardiogram signals in clinical environments. *Engineering Applications of Artificial Intelligence*, 161, Article 112286. <https://doi.org/10.1016/j.engappai.2025.112286>
- Mutlu, E., Hüseyin, V., & Serbes, G. (2024). Bridging auscultation and tiny machine learning: A digital stethoscope leveraging convolutional neural networks on an embedded device for organ sound analysis. *Traitement Du Signal*, 41(5), 2471–2483. <https://doi.org/10.18280/ts.410521>
- Pal, R., & Barney, A. (2024). Automatic breathing phase identification based on the second derivative of the recorded lung sounds. *Biomedical Signal Processing and Control*, 94, Article 106315. <https://doi.org/10.1016/j.bspc.2024.106315>
- Poh, Y. Y., Grooby, E., Tan, K., Malhotra, A., Harandi, M., & Marzbanrad, F. (2026). Reprogramming automatic speech recognition models for neonatal chest sound separation. *IEEE Journal of Biomedical and Health Informatics*, 1–12. <https://doi.org/10.1109/jbhi.2026.3669531>
- Poh, Y. Y., Grooby, E., Tan, K., Zhou, L., King, A., Ramanathan, A., Malhotra, A., Harandi, M., & Marzbanrad, F. (2024). NeoSSNet: Real-time neonatal chest sound separation using deep learning. *IEEE Open Journal of Engineering in Medicine and Biology*, 5, 345–352. <https://doi.org/10.1109/ojemb.2024.3401571>
- Pouyani, M. F., Vali, M., & Ghasemi, M. A. (2022). Lung sound signal denoising using discrete wavelet transform and artificial neural network. *Biomedical Signal Processing and Control*, 72, Article 103329. <https://doi.org/10.1016/j.bspc.2021.103329>
- Reyna, M. A., Kiarashi, Y., Elola, A., Oliveira, J., Renna, F., Gu, A., Perez Alday, E. A., Sadr, N., Sharma, A., Kpodonu, J., Mattos, S., Coimbra, M. T., Sameni, R., Rad,

- A. B., & Clifford, G. D. (2023). Heart murmur detection from phonocardiogram recordings: The George B. Moody PhysioNet Challenge 2022. *PLOS Digital Health*, 2(9), Article e0000324. <https://doi.org/10.1371/journal.pdig.0000324>
- Rocha, B. M., Pessoa, D., Marques, A., Carvalho, P. de, & Paiva, R. P. (2023). Automatic wheeze segmentation using harmonic-percussive source separation and empirical mode decomposition. *IEEE Journal of Biomedical and Health Informatics*, 27(4), 1926–1934. <https://doi.org/10.1109/jbhi.2023.3248265>
- Roy, T. S., Roy, J. K., Mandal, N., & Mukhopadhyay, S. C. (2024). Recent advances in PCG signal analysis using AI: A review. *International Journal on Smart Sensing and Intelligent Systems*, 17(1), Article 20240012. <https://doi.org/10.2478/ijssis-2024-0012>
- Shuvo, S. B., Alam, S. S., & Hasan, T. (2026). A self-attention-driven deep denoiser model for real time lung sound denoising in noisy environments. *Biomedical Signal Processing and Control*, 113, Article 108818. <https://doi.org/10.1016/j.bspc.2025.108818>
- Somarathne, P., Herath, S., Gargiulo, G., Breen, P., Anderson, N., Yao, Y., Liu, T., & Withana, A. (2026). A dual classifier-regressor architecture for heart sound onset/offset detection. *IEEE Transactions on Biomedical Engineering*, 1–9. <https://doi.org/10.1109/tbme.2026.3654558>
- Sun, W., Zhang, Y., & Chen, F. (2024). Research on heart and lung sound separation method based on DAE–NMF–VMD. *EURASIP Journal on Advances in Signal Processing*, 2024(1), Article 59. <https://doi.org/10.1186/s13634-024-01152-0>
- Thinklabs Medical LLC. (n.d.). *Thinklabs Wave app*. Retrieved July 31, 2026, from <https://www.thinklabs.com/thinklabs-wave-app-1>

- Torabi, Y., Shirani, S., & Reilly, J. P. (2023). *A new non-negative matrix factorization approach for blind source separation of cardiovascular and respiratory sound based on the periodicity of heart and lung function* [Preprint]. arXiv.
<https://doi.org/10.48550/arXiv.2305.01889>
- Torabi, Y., Shirani, S., & Reilly, J. P. (2025a). Descriptor: Heart and lung sounds dataset recorded from a clinical manikin using digital stethoscope (HLS-CMDS). *IEEE Data Descriptions*, 2, 133–140. <https://doi.org/10.1109/ieeedata.2025.3566012>
- Torabi, Y., Shirani, S., & Reilly, J. P. (2025b). *Large language model-based nonnegative matrix factorization for cardiorespiratory sound separation* [Preprint]. arXiv.
<https://doi.org/10.48550/arXiv.2502.05757>
- Tsai, T.-S., Hsu, C.-C., Hsieh, C.-L., & Liu, Y.-J. (2023). Separation of heart and lung sounds by a deep network-based model. *2023 IEEE 6th International Conference on Knowledge Innovation and Invention (ICKII)*, 631–633.
<https://doi.org/10.1109/ickii58656.2023.10332660>
- Ullah, R., & Zhang, S. (2024). Unsupervised framework for single channel heart and lung sounds separation in data constrained environments. *Applied Acoustics*, 223, Article 110076. <https://doi.org/10.1016/j.apacoust.2024.110076>
- Wang, W., Qin, D., Wang, S., Fang, Y., & Zheng, Y. (2023). A multi-channel UNet framework based on SNMF-DCNN for robust heart-lung-sound separation. *Computers in Biology and Medicine*, 164, Article 107282.
<https://doi.org/10.1016/j.compbiomed.2023.107282>
- Wang, W., Wang, S., Qin, D., Fang, Y., & Zheng, Y. (2023). Heart-lung sound separation by nonnegative matrix factorization and deep learning. *Biomedical Signal Processing and Control*, 79, Article 104180.
<https://doi.org/10.1016/j.bspc.2022.104180>

- Wang, Y., Yang, H., Gao, Z., Dai, Z., Yu, K., Song, T., Yang, H., Wang, Y., & Zhang, J. (2026). A phase-enhanced neural network with dual-path transformer for single-channel chest sound separation. *IEEE Journal of Biomedical and Health Informatics*, 30(6), 4721–4733. <https://doi.org/10.1109/jbhi.2025.3631643>
- Zhang, B., Wang, J., Ye, Z., Zhou, L., Xiong, D., Wang, X., & Guo, L. (2025). JASSNet: Heart and lung sound separation network based on joint attention mechanism and semi-supervised learning. *Biomedical Signal Processing and Control*, 104, Article 107525. <https://doi.org/10.1016/j.bspc.2025.107525>
- Zhang, L., He, F., Tan, H., Wang, W., & Wang, S. (2026). An efficient lung sound separation algorithm base on GIHO-VMD. *Computer Methods and Programs in Biomedicine*, 273, Article 109124. <https://doi.org/10.1016/j.cmpb.2025.109124>
- Zhao, Q., Geng, S., Wang, B., Sun, Y., Nie, W., Bai, B., Yu, C., Zhang, F., Tang, G., Zhang, D., Zhou, Y., Liu, J., & Hong, S. (2024). Deep learning in heart sound analysis: From techniques to clinical applications. *Health Data Science*, 4, Article 0182. <https://doi.org/10.34133/hds.0182>
- Zheng, J., Wang, Y., Niu, J., Shi, Y., & Xie, F. (2025). Identifying the respiratory sound based on single-channel separation and hyperdimensional computing. *IEEE Sensors Journal*, 25(13), 24626–24633. <https://doi.org/10.1109/jsen.2025.3557909>
- Zheng, R., & Sanei, S. (2024). Separation of cardiopulmonary sound signals for classification of respiratory diseases. *2024 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 1–6. <https://doi.org/10.1109/apsipaasc63619.2025.10848841>

Appendix A: Gantt Chart

Planned FYP1 Report Schedule for the Project under Review. The schedule below shows the general timing of report production for the project under review. It follows the proposed **CPT6314 Teaching Plan T2610** and current report workflow, from FYP1 briefing to submission of report for interim assessment, followed by preparation of oral presentation.

Term 2610 planned schedule based on the CPT6314 Teaching Plan and current report workflow



Note: The chart shows the FYP1 report schedule and preparation period. It does not claim completed FYP2 implementation or testing results.

Appendix B: FYP1 Meeting Logs



CPT6314 Project (FYP1) Meeting Log Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	06/04/2026	Meeting No.:	1	Meeting Mode:	☐ In-Person ☒ Online
Project ID:	985	Project Type:	☐ Research-based ☒ Application-based		
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation				
Student ID:	1221303805	Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN		
Student Programme and Specialisation:	Bachelor of Computer Science (Hons.) / Software Engineering				
Supervisor Name:	Arshad Jamal				
Co-Supervisor Name: (if applicable)	N/A				
Collaborating Company: (if applicable)	N/A				
Company Supervisor Name: (if applicable)	N/A				

1. WORK DONE

(Please write the details of the work done, after the last meeting). Tasks (tick):

☒ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☒ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Confirmed project title, scope, and objectives.
- Set up report repository and literature review folders.
- Prepared paper screening and PRISMA tracking files.

2. WORK TO BE DONE

(Please write the details of the work to be done, before the next meeting). Tasks (tick):

☒ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☒ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Collect and organize related papers.
- Define inclusion and exclusion criteria.
- Start screening existing papers.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

(Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions)

Some papers focus on disease diagnosis instead of sound separation.
Solution: Limit the review to cardiopulmonary sound separation only.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

☐ Not Satisfactory ☒ Satisfactory

The student has completed the initial project setup phase, including: 1. Confirmation of project title, scope, and objectives. 2. Establishment of report repository and literature review organization. 3. Preparation of paper screening and PRISMA tracking files. Planned next activities include: 1. Collecting and organizing related research papers. 2. Defining inclusion and exclusion criteria. 3. Beginning systematic screening of literature.

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature (if applicable)

Company Supervisor's Signature (if applicable)

IMPORTANT NOTES TO STUDENTS:

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report. Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee. This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.



CPT6314 Project (FYP1) Meeting Log Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	27/04/2026	Meeting No.:	2	Meeting Mode:	☐ In-Person ☒ Online
Project ID:	985	Project Type:	☐ Research-based ☒ Application-based		
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation				
Student ID:	1221303805	Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN		
Student Programme and Specialisation:	Bachelor of Computer Science (Hons.) / Software Engineering				
Supervisor Name:	Arshad Jamal				
Co-Supervisor Name: (if applicable)	N/A				
Collaborating Company: (if applicable)	N/A				
Company Supervisor Name: (if applicable)	N/A				

1. WORK DONE

(Please write the details of the work done, after the last meeting). Tasks (tick):

☒ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Screened papers from 2022–2026.
- Recorded Include, Maybe, and Exclude decisions.
- Updated references, extraction matrix, and PRISMA counts.

2. WORK TO BE DONE

(Please write the details of the work to be done, before the next meeting). Tasks (tick):

☐ Background study	☐ Implementation	☐ Testing	☒ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Download accessible papers using Zotero.
- Audit newly downloaded PDFs.
- Prepare Chapter 2 evidence map.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

(Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions)

Some papers were not accessible through university access.
Solution: Marked them as excluded and recorded the reasons for PRISMA.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

☐ Not Satisfactory ☒ Satisfactory

Download accessible papers and audit the downloaded research papers. Prepare extensive evidence in Chapter 2.

Student's Signature

Supervisor's Signature

Co-Supervisor's Signature (if applicable)

Company Supervisor's Signature (if applicable)

IMPORTANT NOTES TO STUDENTS:

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report. Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee. This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Project (FYP1) Meeting Log

Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	18/05/2026	Meeting No.:	3	Meeting Mode:	☐ In-Person ☒ Online
Project ID:	985	Project Type:	☐ Research-based ☒ Application-based		
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation				
Student ID:	1221303805	Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN		
Student Programme and Specialisation:	Bachelor of Computer Science (Hons.) / Software Engineering				
Supervisor Name:	Arshad Jamal				
Co-Supervisor Name: (if applicable)	N/A				
Collaborating Company: (if applicable)	N/A				
Company Supervisor Name: (if applicable)	N/A				

1. WORK DONE
[Please write the details of the work done, after the last meeting]. Tasks (tick):

☒ Background study	☐ Implementation	☐ Testing	☒ Evaluation
☐ Commercialisation Proposal	☒ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Finalized Chapter 2 evidence map.
- Drafted Chapter 1, Chapter 2, and Chapter 3.
- Generated the report in Word format using Quarto.

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]. Tasks (tick):

☐ Background study	☒ Implementation	☐ Testing	☒ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Add literature review matrix in Chapter 2.
- Number the problem statements and align them with objectives.
- Fix report formatting and front matter.
- Insert the final PRISMA diagram into Chapter 2.
- Start Chapter 4: system design and implementation.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

The report still needs clearer alignment between the problem statements and project objectives. Chapter 2 also needs a literature review matrix to summarize the reviewed papers clearly.
Solution: Revise Chapter 1 by numbering the problems and matching them with the objectives. Update Chapter 2 by adding the literature review matrix based on the selected papers.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)
☐ Not Satisfactory ☒ Satisfactory

Align the problem statements and objectives. Integrate final Prisma diagram.
Prepare a literature review paper in MDPI format.

Student's Signature



Supervisor's Signature



Arshad Jamal
Lecturer
Faculty of Computing and Informatics
Multimedia University, Persiaran Multimedia
63100 Cyberjaya, Selangor Darul Ehsan

Co-Supervisor's
Signature (if applicable)Company Supervisor's
Signature (if applicable)**IMPORTANT NOTES TO STUDENTS:**

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report. Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee. This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Project (FYP1) Meeting Log

Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	27/05/2026	Meeting No.:	4	Meeting Mode:	☐ In-Person ☒ Online
Project ID:	985	Project Type:	☐ Research-based ☒ Application-based		
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation				
Student ID:		Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN		
Student Programme and Specialisation:	Bachelor of Computer Science (Hons.) / Software Engineering				
Supervisor Name:	Arshad Jamal				
Co-Supervisor Name: (if applicable)	N/A				
Collaborating Company: (if applicable)	N/A				
Company Supervisor Name: (if applicable)	N/A				

1. WORK DONE
[Please write the details of the work done, after the last meeting]. Tasks (tick):

☐ Background study	☒ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☒ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Revised the problem statements and objectives.
- Updated the literature review and matrix.
- Inserted the PRISMA screening figure.
- Refactored the report to match the application-based format.
- Started converting Chapter 3 into Requirements Analysis.

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]. Tasks (tick):

☐ Background study	☒ Implementation	☐ Testing	☒ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Complete Chapter 3 Requirements Analysis.
- Prepare the Google Form questionnaire.
- Develop Chapter 4 system design diagrams.
- Define the main prototype functions.
- Check formatting, citations, and appendices.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

The report structure did not fully match the application-based FYP format.

Solution: The report was reorganized into Requirements Analysis, System Design, Implementation Plan, and Conclusion.

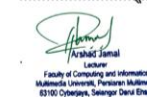
4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)
☐ Not Satisfactory ☒ Satisfactory

Steady progress being made. Need to list all the functional and non functional requirements in detail to draft the chapter 3.

Student's Signature



Supervisor's Signature



Arshad Jamal
Lecturer
Faculty of Computing and Informatics
Multimedia University, Persiaran Multimedia
63100 Cyberjaya, Selangor Darul Ehsan

Co-Supervisor's
Signature (if applicable)Company Supervisor's
Signature (if applicable)**IMPORTANT NOTES TO STUDENTS:**

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report. Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee. This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Project (FYP1) Meeting Log
Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	10/06/2026	Meeting No.:	5	Meeting Mode:	☐ In-Person ☒ Online
Project ID:	985	Project Type:	☐ Research-based ☒ Application-based		
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation				
Student ID:		Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN		
Student Programme and Specialisation:	Bachelor of Computer Science (Hons.) / Software Engineering				
Supervisor Name:	Arshad Jamal				
Co-Supervisor Name: (if applicable)	N/A				
Collaborating Company: (if applicable)	N/A				
Company Supervisor Name: (if applicable)	N/A				

1. WORK DONE
[Please write the details of the work done, after the last meeting]. Tasks (tick):

☐ Background study	☒ Implementation	☐ Testing	☒ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Completed the Chapter 3 questionnaire template.
- Prepared the Google Form guide.
- Added placeholder figures for pending responses.
- Completed the main Chapter 4 system diagrams.
- Added model selection to the system design.

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]. Tasks (tick):

☐ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Expand Chapter 5 Implementation Plan.
- Describe the planned separation algorithms.
- Add evaluation metric explanations.
- Validate and regenerate all diagrams.
- Render and check the final DOCX report.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

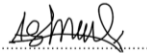
Questionnaire responses have not been collected yet.

Solution: Placeholder figures were used, and findings were marked as pending until real responses are collected.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

☐ Not Satisfactory ☒ Satisfactory

Student's Signature



Supervisor's Signature



Arshad Jamal
Lecturer
Faculty of Computing and Informatics
Multimedia University, Persiaran Multimedia
63100 Cyberjaya, Selangor Darul Ehsan

Co-Supervisor's
Signature (if applicable)Company Supervisor's
Signature (if applicable)**IMPORTANT NOTES TO STUDENTS:**

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report. Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee. This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

CPT6314 Project (FYP1) Meeting Log
Trimester: MARCH / APRIL 2026 (Trimester ID:2610)

Meeting Date:	24/06/2026	Meeting No.:	6	Meeting Mode:	☐ In-Person ☒ Online
Project ID:	985	Project Type:	☐ Research-based ☒ Application-based		
Project Title:	Machine Learning-Based System for Cardiopulmonary Sound Separation				
Student ID:		Student Name:	AL-SALOUL, ASHRAF ALI HUSSEIN		
Student Programme and Specialisation:	Bachelor of Computer Science (Hons.) / Software Engineering				
Supervisor Name:	Arshad Jamal				
Co-Supervisor Name: (if applicable)	N/A				
Collaborating Company: (if applicable)	N/A				
Company Supervisor Name: (if applicable)	N/A				

1. WORK DONE
[Please write the details of the work done, after the last meeting]. Tasks (tick):

☐ Background study	☒ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Completed Chapter 5 Implementation Plan.
- Added planned algorithm and metric explanations.
- Added Chapter 6 Conclusion.
- Updated the system design diagrams.
- Regenerated and validated the DOCX report.

2. WORK TO BE DONE
[Please write the details of the work to be done, before the next meeting]. Tasks (tick):

☐ Background study	☐ Implementation	☐ Testing	☐ Evaluation
☐ Commercialisation Proposal	☐ Research Paper	☐ Draft Report	☐ Final Report

Details (max 3-5 bullet points):

- Update all fields in Microsoft Word.
- Check TOC, figures, tables, and page numbers.
- Insert the required appendices.
- Confirm final scope with the supervisor.
- Collect Google Form responses later.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS
[Please write the details of the problems encountered, after the last meeting and provide the solutions / plan for the solutions]

The report still needs manual checking before submission.

Solution: The DOCX will be checked in Word, and pending questionnaire findings will be updated after real responses are collected.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

☐ Not Satisfactory ☒ Satisfactory

Student's Signature



Supervisor's Signature



Arshad Jamal
Lecturer
Faculty of Computing and Informatics
Multimedia University, Persiaran Multimedia
63100 Cyberjaya, Selangor Darul Ehsan

Co-Supervisor's
Signature (if applicable)Company Supervisor's
Signature (if applicable)**IMPORTANT NOTES TO STUDENTS:**

- Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- Student must upload the soft copies of the meeting logs and attach them along with final (FYP2) report. Minimum requirement is SIX Meeting Logs (Period: Week 4 to Week 14). Students can have fortnightly meetings with the supervisor.
- Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and for checking the attendance requirement of the student, by the FYP Committee. This also provides the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provides the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
- Student who fails to meet the minimum requirement (six nos.) of log sheets will not be allowed to submit FYP report.

Appendix C: Turnitin Similarity Index Page

ALSALOUL ASHRAF ALI HUSSEIN - 1221303805_Arshad_Jamal_FYP1_Report			
ORIGINALITY REPORT			
16%	12%	12%	9%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS
PRIMARY SOURCES			
1	Submitted to Multimedia University Student Paper		
2	arxiv.org Internet Source		
3	sameni.info Internet Source		
4	documentsdelivered.com Internet Source		
5	Yang Yi Poh, Ethan Grooby, Kenneth Tan, Atul Malhotra, Mehrtash Harandi, Faezeh Marzbanrad. "Reprogramming Automatic Speech Recognition Models for Neonatal Che Sound Separation", IEEE Journal of Biomedical and Health Informatics, 2026 Publication		
6	Shams Nafisa Ali, Afia Zahin, Samiul Based Shuvo, Nusrat Binta Nizam et al. "BUET multi disease heart sound dataset: A comprehensive auscultation dataset for developing computer-aided diagnostic systems", Computer Methods and Programs in Biomedicine Update, 2026 Publication		
7	research.monash.edu Internet Source		
8	acikerisim.harran.edu.tr:8080 Internet Source		
9	digitalcommons.odu.edu Internet Source		
10	pubmed.ncbi.nlm.nih.gov Internet Source		
11	Leiming Zhang, Fuliang He, Hao Tan, Wangquan Wang, Shiyuan Wang. "An efficient lung sound separation algorithm base on GIHO-VMD", Computer Methods and Programs in Biomedicine, 2026 Publication		
12	Submitted to Aalto Yliopisto Student Paper		

13	ouci.dntb.gov.ua Internet Source
14	etd.repository.ugm.ac.id Internet Source
15	Jaspreet Kaur, Brijesh Kumar. "Comparative Analysis of Machine Learning Algorithms for Lung Sound Analysis for Disease Prediction", 2025 3rd International Conference on Advances in Computation, Communication and Information Technology (ICAICIT), 2025 Publication
16	Submitted to University of North Texas Student Paper
17	vbn.aau.dk Internet Source
18	Jing Lyu, Sijie Su, Xin Wang, Enhai Song, Qinyuan Zhang. " High Performance Flexible Optical Fiber Sensor Enabled by Er /Yb Co-Doped near Pure-Green Upconversion Emission Phosphor for Implantable Temperature Sensing ", Advanced Materials Interfaces, 2026 Publication
19	research.buaa.edu.cn Internet Source
20	www.mdpi.com Internet Source
21	data.mendeley.com Internet Source
22	Wenhui Sun, Yipeng Zhang, Fuming Chen. "Research on heart and lung sound separation method based on DAE-NMF-VMD", EURASIP Journal on Advances in Signal Processing, 2024 Publication
23	Daisuke Niizumi, Daiki Takeuchi, Masahiro Yasuda, Binh Thien Nguyen, Yasunori Ohishi Noboru Harada. "Assessing the Utility of Audio Foundation Models for Heart and Respiratory Sound Analysis", 2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), 2025 Publication
24	Submitted to University of South Wales - Pontypridd and Cardiff Student Paper
25	Maros Jakubec, Eva Lieskovska, Peter Pocka. "A robust deep learning based model for denoising phonocardiogram signals in clinical environments", Engineering Applications of Artificial Intelligence, 2025 Publication

- 26 hdl.handle.net
Internet Source
- 27 Francis Chikweto, Muyanga Kampekete, Mubanga Boyd, Kalupa Conix, Munsanje Foste Chilufya Queen. "Toward the Development of a Low-Cost AI-Enabled Digital Stethoscop 2024 International Conference on Electrical and Computer Engineering Researches (ICECER), 2024
Publication
- 28 Submitted to Nightingale College - School of Nursing
Student Paper
- 29 Peng-Fei Zhang, Hua-Nong Ting, Siow-Wee Chang. "Using Cough Sounds for the Recognition of Multiclass Respiratory Diseases With Artificial Intelligence: A Review", IEEE Access, 2025
Publication
- 30 Van-Thuan Tran, Ting-Hao You, Wei-Ho Tsai. "Acoustic Cues for Person Identification Using Cough Sounds", Computer Methods and Programs in Biomedicine Update, 2025
Publication
- 31 Alessandra Fava, Behnood Dianat, Alessandro Bertacchini, Andreina Manfredi, Marco Sebastiani, Marco Modena, Fabrizio Pancaldi. "Pre-processing techniques to enhance the classification of lung sounds based on deep learning", Biomedical Signal Processing and Control, 2024
Publication
- 32 Submitted to University Of Tasmania
Student Paper
- 33 bura.brunel.ac.uk
Internet Source
- 34 sciendo.com
Internet Source
- 35 Submitted to Nanyang Technological University
Student Paper
- 36 open.library.ubc.ca
Internet Source
- 37 Ming Hung Wong. "Handbook of Microplastics and Associated Chemicals - Ecological and Health Impacts", CRC Press, 2026
Publication
- 38 Xu Wang, Hui Zhou, Yanjie Xu, Yingtao Zhang, Zhenguo Nie. "Multimodal Acoustic Feature Fusion with Hybrid Deep Networks for Industrial Equipment Fault Detection", 2025 International Conference on Equipment Intelligent Operation and Maintenance (ICEION 2025)

- Publication
- 39 nitrkl.ac.in
Internet Source
- 40 Chenwen Wu, Na Ye, Jialin Jiang. "Classification and Recognition of Lung Sounds Based Improved Bi-ResNet Model", IEEE Access, 2024
Publication
- 41 doaj.org
Internet Source
- 42 Bowen Han, Wei Quan. "Cardiorespiratory Sound Separation Using Singular Spectrum Analysis", 2025 17th International Conference on Signal Processing Systems (ICSPS), 2025
Publication
- 43 Malaya Kumar Nath, Sanghamitra Subhadarsini Dash, Karthick Mookkandi, Thivya A.. "Towards intelligent phonocardiogram analysis: Advances and future perspectives in cardiovascular disease detection", Engineering Applications of Artificial Intelligence, 2025
Publication
- 44 Rizwan Ullah, Shaohui Zhang. "Unsupervised framework for single channel heart and lung sounds separation in data constrained environments", Applied Acoustics, 2024
Publication
- 45 iieta.org
Internet Source
- 46 Submitted to Faculty of Medicine & Health Sciences
Student Paper
- 47 Subhasikta Behera, Puneet Kumar Jain, Sambit Bakshi. "Adaptive thresholding of DWT coefficients using UNet for denoising real-life respiratory sounds", Biomedical Signal Processing and Control, 2026
Publication
- 48 "Table of Contents", IEEE Journal of Biomedical and Health Informatics, 2026
Publication
- 49 Wangquan Wang, Chenghong Bo, An Zhao, Leiming Zhang, Fuliang He. "Heart-lung sound unsupervised separation by nonnegative matrix factorization and modified Hilbert spectrum", Biomedical Signal Processing and Control, 2025
Publication
- 50 assets-eu.researchsquare.com
Internet Source
- 51 www.newbornjournal.org
Internet Source
- 52 Submitted to University of Stellenbosch, South Africa
Student Paper

- 53 www.xiaolei-zhang.net
Internet Source
- 54 Bochao Zhang, Jiping Wang, Zhipeng Ye, Linfu Zhou, Daxi Xiong, Xiaojun Wang, Liquan Guo. "JASSNet: Heart and lung sound separation network based on joint attention mechanism and Semi-Supervised learning", Biomedical Signal Processing and Control, 2025
Publication
- 55 Submitted to University of Pretoria
Student Paper
- 56 Yanshan Sha, Dongliang Chen, Feiyang Wu, Qiang Cao, Yucong Zhou, Weihua Zhang, Xi Huang, Minghui Wang. "Experimental investigation of time-frequency characteristics of acoustic signals during boilover in small-scale oil tanks", Journal of Loss Prevention in the Process Industries, 2026
Publication
- 57 erepo.uef.fi
Internet Source
- 58 jeeemi.org
Internet Source
- 59 suraj.lums.edu.pk
Internet Source
- 60 H.L. Gururaj, Francesco Flammini, S. Srividhya, M.L. Chayadevi, Sheba Selvam. "Computational Science Engineering", CRC Press, 2024
Publication
- 61 Yuqi Wang, Han Yang, Zhixing Gao, Zhiwei Dai, Kang Yu, Tingting Song, Hao Yang, Yunfei Wang, Jun Zhang. "A Phase-Enhanced Neural Network With Dual-Path Transformer for Single-Channel Chest Sound Separation", IEEE Journal of Biomedical and Health Informatics, 2026
Publication
- 62 digibuo.uniovi.es
Internet Source
- 63 ds.inflibnet.ac.in
Internet Source
- 64 "Swarm, Evolutionary, and Memetic Computing", Springer Science and Business Media LLC, 2015
Publication
- 65 Ayman Anwar, Yassin Khalifa, James L. Coyle, Ervin Sejdic. "Transformers in biosignal analysis: A review", Information Fusion, 2025
Publication

- 66 Ravi Pal, Anna Barney. "Automatic breathing phase identification based on the second derivative of the recorded lung sounds", Biomedical Signal Processing and Control, 2025
Publication
- 67 github.com
Internet Source
- 68 "2023 Index IEEE Journal of Biomedical and Health Informatics Vol. 27", IEEE Journal of Biomedical and Health Informatics, 2023
Publication
- 69 www.papers.phmsociety.org
Internet Source
- 70 "Applications of Computational Intelligence in Management & Mathematics", Springer Science and Business Media LLC, 2023
Publication
- 71 E. Grooby, C. Sitaula, D. Fattahi, R. Sameni, K. Tan, L. Zhou, A. King, A. Ramanathan, A. Malhotra, G.A. Dumont, F. Marzbanrad. "Noisy Neonatal Chest Sound Separation for High Quality Heart and Lung Sounds", IEEE Journal of Biomedical and Health Informatics, 2025
Publication
- 72 researchr.org
Internet Source
- 73 www.peeref.com
Internet Source
- 74 Submitted to British University in Egypt
Student Paper
- 75 Lok, Richard Him. "Tool Support for Software Process Assessments", University of the Witwatersrand, Johannesburg (South Africa), 2025
Publication
- 76 Weibo Wang, Shubo Wang, Dimei Qin, Yu Fang, Yongkang Zheng. "Heart-lung sound separation by nonnegative matrix factorization and deep learning", Biomedical Signal Processing and Control, 2023
Publication
- 77 Yasaman Torabi, Shahram Shirani, James P. Reilly. "LingoNMF: Large language model-enhanced nonnegative matrix factorization for cardiorespiratory sound separation", Biomedical Signal Processing and Control, 2026
Publication
- 78 assets.researchsquare.com
Internet Source
- 79 online-journals.org
Internet Source

80 portalcris.lsmuni.lt
Internet Source

81 scholarshare.temple.edu
Internet Source

82 Submitted to upGrad
Student Paper

83 Ethan Grooby, Jinyuan He, Julie Kiewsky, Davood Fattahi et al. "Neonatal Heart and Lung Sound Quality Assessment for Robust Heart and Breathing Rate Estimation for telehealth Applications", IEEE Journal of Biomedical and Health Informatics, 2020
Publication

84 Submitted to Sunway Education Group
Student Paper

85 medinform.jmir.org
Internet Source

86 mobt3ath.com
Internet Source

87 Meltem Çınar Bozdağ, Alper Kumcu, Lütfi Kerem Şenel, Havva Nur Temizkan et al. "Spor language biomarkers in Turkish-speaking schizophrenia patients: Evidence from linguistic analysis and word embeddings", Psychiatry Research, 2026
Publication

88 Submitted to Sheffield Hallam University
Student Paper

89 ijmada.com
Internet Source

90 os.unil.cloud.switch.ch
Internet Source

91 pdfs.semanticscholar.org
Internet Source

92 researchbank.swinburne.edu.au
Internet Source

93 scholar.hit.edu.cn
Internet Source

94 Arshad Jamal, R. Kanesaraj Ramasamy, Junaidi Abdullah, Sivasutha Thanjappan, Faizal Amri Hamzah, Shamsuriani Bt Md Jamal. "Chapter 4 Energy-Efficient Biomedical Signal

Decoupling Via Optimized Generative AI Models", Springer Science and Business Media LLC, 2026
Publication

95 Submitted to University of the West Indies
Student Paper

96 ase.in.tum.de
Internet Source

97 baltijapublishing.lv
Internet Source

98 cdn.techscience.cn
Internet Source

99 ebin.pub
Internet Source

100 researchbank.rmit.edu.au
Internet Source

101 www.coursehero.com
Internet Source

102 Ali Hussein Shaker, Ibrahim Amer Ibrahim, Sadik Kamel Gharghan. "Machine learning techniques for cardiovascular disease detection through heart sound analysis: A review AIP Publishing, 2024
Publication

103 Arvind Dagur, Dharendra Kumar Shukla, Nazarov Fayzullo Makhmadiyrovich, Akhatov Akmal Rustamovich, Jabborov Jamol Sindorovich. "Artificial Intelligence and Information Technologies", CRC Press, 2024
Publication

104 Submitted to Capella University
Student Paper

105 Green, Virginia. "Entrepreneurial boot camps: A case study to evaluate and improve the establishment of methods, goals, strategies and tasks", Proquest, 20111109
Publication

106 K. Sathesh, P Kowsalya, E. Aravindraj, S.Guru Naveen Reddy, K. Anvith Sai Varma, P. Ganesh. "Clearer HeartBeats: Enhancement of cardiac sounds using Adaptive Filtering, Wavelet Decomposition", 2024 10th International Conference on Communication and Signal Processing (ICCSP), 2024
Publication

107 Submitted to Midlands State University
Student Paper

108

R. N. V. Jagan Mohan, B. H. V. S. Rama Krishnam Raju, V. Chandra Sekhar, T. V. K. P. Prasanna. "Algorithms in Advanced Artificial Intelligence - Proceedings of International Conference on Algorithms in Advanced Artificial Intelligence (ICAAL-2024)", CRC Press, 2025

Publication

109

Yuan Xie, Kan Xie, Qiyu Yang, Shengli Xie. "Reverberant blind separation of heart and lung sounds using nonnegative matrix factorization and auxiliary function technique", Biomedical Signal Processing and Control, 2021

Publication

110

eprints.bournemouth.ac.uk

Internet Source

111

eprints.utar.edu.my

Internet Source

112

es.scribd.com

Internet Source

113

etd.uum.edu.my

Internet Source

114

journal.50sea.com

Internet Source

115

journals.plos.org

Internet Source

116

knowledge.lancashire.ac.uk

Internet Source

117

par.nsf.gov

Internet Source

118

research.utwente.nl

Internet Source

119

studentsrepo.um.edu.my

Internet Source

120

www.csauthors.net

Internet Source

121

Adekanmbi, Motunrayo Helen. "Knowledge of Tuberculosis Patients About Their Diseases in Tshwane, South Africa", University of South Africa (South Africa)

Publication

122

Chunjian Yang, Nan Hu, Dongyang Xu, Zhi Wang, Shengsheng Cai. "Monaural cardiopulmonary sound separation via complex-valued deep autoencoder and cyclostationarity", Biomedical Physics & Engineering Express, 2023

Publication

123

Mahmoud Fakhry, Abeer FathAllah Brery. "Generative artificial intelligence for heart sound analysis: From model architectures to clinical applications", Intelligent Systems with Applications, 2026

Publication

124

Przemyslaw Pietrzak, Marcin Wolkiewicz. "Demagnetization Fault Diagnosis of Permanent Magnet Synchronous Motors Based on Stator Current Signal Processing and Machine Learning Algorithms", Sensors, 2023

Publication

125

Shuvashis Sarker, Faika Fairuj Preotee, Tashreef Muhammad, Shamim Akhter. "An Innovative Framework for Heart Sound Classification Integrating Adaptive Fuzzy Rank-Based Ensemble of Transfer Learning Models with Multi-Dimensional Features", Cold Spring Harbor Laboratory, 2025

Publication

126

Yin Liang, Xiaoying Yang, Runrui Han, Yingchen Jia, Gaoxu Xu. "A learnable multi-order spatial-temporal attention network for fMRI-based neurological disorder identification", Biomedical Signal Processing and Control, 2026

Publication

Exclude quotes	Off	Exclude matches	Off
Exclude bibliography	Off		

Appendix D: Full Literature Review Matrix

The whole literature review matrix below provides an opportunity to assess the sources included in the chapter 2 synthesis. It allows for comparing how each of the cited works addressed different research aims, approaches, data sets, results, limitations, and applicability to the proposed cardiopulmonary sound separation system.

No.	Research Title	Author(s) & Year	Journal / Conference / Database	Research Problem / Objective	Algorithm / Framework Used	Dataset Used	Results / Findings	Research Gap	Limitation	How It Helps This Project
1	An End-to-End Deep Learning Framework for Real-Time Denoising of Heart Sounds for Cardiac Disease Detection in Unseen Noise	Ali et al. (2023) (Ali et al., 2023)	IEEE Access	Improve noisy heart sound recordings before later analysis.	LU-Net encoder-decoder denoiser.	PCG data mixed with breathing and hospital noise.	Reported SNR and real-time denoising improvements.	Denoising is related but does not separate heart and lung outputs.	Focuses on heart sound denoising and diagnosis support.	Supports preprocessing and noise-reduction choices.
2	Noisy Neonatal Chest Sound Separation for High-Quality Heart and Lung Sounds	Grooby et al. (2023) (Grooby et al., 2023)	IEEE Journal of Biomedical and Health Informatics	Separate noisy neonatal chest recordings into heart and lung sounds.	NMF and NMCF.	Neonatal artificial mixtures and noisy chest sounds.	Improved signal quality and vital-sign estimation support.	Robust separation remains difficult under noisy neonatal conditions.	Dataset is neonatal and not clearly public.	Provides a classical separation baseline.
3	NeoSSNet: Real-Time Neonatal Chest Sound Separation Using Deep Learning	Poh et al. (2024) (Poh et al., 2024)	IEEE Open Journal of Engineering in Medicine and Biology	Perform real-time neonatal chest sound separation.	1D encoder-decoder with transformer mask generation.	Neonatal chest recordings with heart and lung references.	Reported SDR and SI-SDR improvement with real-time relevance.	Reproducibility is limited by private clinical data.	Performance depends on noise and recording quality.	Main deep learning reference for the prototype.
4	Recent Advances in PCG Signal Analysis using AI: A Review	Roy et al. (2024) (Roy et al., 2024)	International Journal on Smart Sensing and Intelligent Systems	Review AI methods for PCG preprocessing and analysis.	Review of preprocessing, segmentation, features, and AI models.	PCG datasets from prior studies.	Summarizes common PCG processing stages and AI approaches.	Review is not focused on heart-lung source separation.	Mainly classification and PCG analysis oriented.	Supports background on PCG preprocessing.
5	Research on Heart and Lung Sound Separation	Sun et al. (2024) (Sun et al., 2024)	EURASIP Journal on Advances in	Improve heart-lung sound separation using	Deep autoencoder, NMF, and VMD.	Heart-lung sound mixtures.	Reported SDR, PESQ, STOI, and	Reproducible public data access is limited.	Data and code access may require request.	Shows how ML and decomposition

	Method Based on DAE-NMF-VMD		Signal Processing	a hybrid method.			separation metrics.			can be combined.
6	Unsupervised Framework for Single Channel Heart and Lung Sounds Separation in Data Constrained Environments	Ullah and Zhang (2024) (Ullah and Zhang, 2024)	Applied Acoustics	Separate single-channel heart and lung sounds with limited data.	MCRA, OMLSA, and NMF-HMM.	Heart and lung sound mixtures.	Reported SDR, SAR, and separation performance.	Data-constrained separation needs methods with limited labels.	Dataset availability is unclear.	Supports feasible low-data separation options.
7	A Multi-Channel UNet Framework Based on SNMF-DCNN for Robust Heart-Lung-Sound Separation	Wang et al. (2023) (Wang et al., 2023a)	Computers in Biology and Medicine	Improve robust heart-lung sound separation.	SNMF-DCNN-MCUNet.	Heart-lung mixtures with different lung sound types.	Reported SNR, SDR, SIR, SAR, and correlation improvements.	Constructed mixtures affect generalizability.	Public dataset details are limited.	Guides U-Net-style model design and metrics.
8	Heart-Lung Sound Separation by Nonnegative Matrix Factorization and Deep Learning	Wang et al. (2023) (Wang et al., 2023b)	Biomedical Signal Processing and Control	Combine NMF and deep learning for separation.	MCNMF, clustering, and ECNet.	Heart-lung sound mixtures.	Reported SNR, SDR, SIR, SAR, and correlation.	Separation depends on mixture assumptions.	Public dataset details are limited.	Supports hybrid baseline and evaluation planning.
9	JASSNet: Heart and Lung Sound Separation Network Based on Joint Attention Mechanism and Semi-Supervised Learning	Zhang et al. (2025) (Zhang et al., 2025)	Biomedical Signal Processing and Control	Improve separation using attention and semi-supervision.	Joint attention and semi-supervised network.	Simulated and clinical cardiopulmonary recordings.	Reported SDR, SIR, SAR, and rate-estimation improvement.	Advanced models can be difficult to reproduce.	Clinical data and disease context reduce general reuse.	Defines a recent advanced comparison method.
10	Deep Learning in Heart Sound Analysis: From Techniques to Clinical Applications	Zhao et al. (2024) (Zhao et al., 2024)	Health Data Science	Review deep learning for heart sound analysis.	Review of CNN, RNN, transformer, denoising, and segmentation methods.	Common heart sound datasets.	Summarizes heart sound AI methods and dataset issues.	Not focused on heart-lung source separation.	Clinical analysis focus may not match this project scope.	Supports ML background and dataset discussion.
11	Separation of Cardiopulmonary Sound Signals for Classification of	Zheng and Sanei (2024) (Zheng and Sanei, 2024)	APSIPA ASC	Separate cardiopulmonary signals before	Singular spectrum analysis with	Auscultated cardiopulmonary recordings.	Separation improved downstream	Separation quality is secondary to classification.	Diagnostic objective differs from this FYP.	Shows separation value without shifting project

	Respiratory Diseases			respiratory analysis.	kurtosis optimization.		respiratory classification.			scope to diagnosis.
12	Reprogramming Automatic Speech Recognition Models for Neonatal Chest Sound Separation	Poh et al. (2026) (Poh et al., 2026)	IEEE Journal of Biomedical and Health Informatics	Adapt speech models for neonatal chest sound separation.	Whisper model reprogramming.	Neonatal chest mixtures and supporting PCG data.	Reported separation, segmentation, and rate-estimation results.	Foundation-model adaptation is complex.	Small neonatal source set and artificial mixtures limit generalization.	Identifies an advanced ML direction beyond the first prototype.
13	A Phase-Enhanced Neural Network With Dual-Path Transformer for Single-Channel Chest Sound Separation	Wang et al. (2026) (Wang et al., 2026)	IEEE Journal of Biomedical and Health Informatics	Improve single-channel chest sound reconstruction.	Phase-enhanced dual-path transformer.	Constructed heart and lung sound mixtures.	Reported SI-SDR-based and phase-aware improvements.	Phase reconstruction is often under-addressed.	Relies on constructed mixtures.	Supports time-frequency and reconstruction design choices.
14	Large Language Model-based Nonnegative Matrix Factorization For Cardiorespiratory Sound Separation	Torabi et al. (2025) (Torabi et al., 2025b)	arXiv	Use language-model feedback to tune NMF separation.	LingoNMF with LLaMA-based feedback.	Synthetic and manikin heart-lung recordings.	Reported SNR, SDR, SIR, and SAR improvements.	LLM-assisted separation adds system complexity.	Preprint and synthetic/manikin data limit claims.	Helps compare explainable NMF extensions.
15	A New Non-Negative Matrix Factorization Approach for Blind Source Separation of Cardiovascular and Respiratory Sound Based on the Periodicity of Heart and Lung Function	Torabi et al. (2023) (Torabi et al., 2023)	arXiv	Use periodicity to separate cardiovascular and respiratory sounds.	Periodicity-guided NMF.	Synthetic heart and lung sound mixtures.	Reported SDR, SIR, and SAR.	Real-world validation remains limited.	Mainly synthetic evaluation.	Supports physiological periodicity as a separation cue.
16	Separation of Heart and Lung Sounds by A	Tsai et al. (2023) (Tsai et al., 2023)	International Conference on Knowledge	Separate generated heart-lung mixtures	Deep network-based model.	Public heart and lung sounds used to	Reported correlation-based	Evaluation needs stronger metrics and real recordings.	Synthetic mixtures and limited method detail.	Provides a feasible student-level ML reference.

	Deep Network-Based Model		Innovation and Invention	using a deep model.		generate mixtures.	separation results.			
17	An efficient lung sound separation algorithm base on GIHO-VMD	Zhang et al. (2026) (Zhang et al., 2026)	Computer Methods and Programs in Biomedicine	Separate lung sounds from mixed heart-lung signals.	GIHO-VMD.	Pre-simulated mixed heart and lung sound signals.	Reported SNR, NCC, spectrogram, and runtime comparisons.	Decomposition methods remain sensitive to simulation assumptions.	Real-world application pipeline is not fully shown.	Provides a recent decomposition comparator.
18	Identifying the Respiratory Sound Based on Single-Channel Separation and Hyperdimensional Computing	Zheng et al. (2025) (Zheng et al., 2025)	IEEE Sensors Journal	Use separation before respiratory sound identification.	Encoder-decoder separation, hyperdimensional computing, and LightGBM.	ICU respiratory recordings and benchmark data.	Reported respiratory identification improvements.	Separation is used mainly for classification.	Standalone separation quality is not the main objective.	Shows how separation can support later audio analysis.
19	Descriptor: Heart and Lung Sounds Dataset Recorded from a Clinical Manikin using Digital Stethoscope (HLS-CMDS)	Torabi et al. (2025) (Torabi et al., 2025a)	IEEE Data Descriptions	Provide paired recordings for separation research.	Dataset descriptor.	HLS-CMDS manikin heart, lung, mixed, and source recordings.	Provides mixed recordings with corresponding source signals.	Public paired source datasets remain limited.	Manikin data may not represent full patient variability.	Main public dataset candidate for training and evaluation.
20	Automatic Wheeze Segmentation Using Harmonic-Percussive Source Separation and Empirical Mode Decomposition	Rocha et al. (2023) (Rocha et al., 2023)	IEEE Journal of Biomedical and Health Informatics	Segment wheezes from respiratory recordings.	HPSS and EMD.	Respiratory sound datasets.	Reported wheeze segmentation performance.	Respiratory preprocessing is not heart-lung separation.	Full real-world wheeze segmentation remains difficult.	Supports preprocessing and respiratory signal handling.
21	A Blind Filtering Framework for Noisy Neonatal Chest Sounds	Fattahi et al. (2022) (Fattahi et al., 2022)	IEEE Access	Filter noisy neonatal chest sounds.	Wavelet transform, PCA, periodic component analysis, and SOBI.	Neonatal chest sound recordings.	Improved quality assessment and rate estimation.	Traditional filtering needs careful component selection.	Severe artefacts may still reduce reliability.	Supports preprocessing and baseline separation design.

22	A Dual Classifier-Regressor Architecture for Heart Sound Onset/Offset Detection	Somarathne et al. (2026) (Somarathne et al., 2026)	IEEE Transactions on Biomedical Engineering	Detect heart sound onset and offset.	Classifier-regressor PCG architecture.	PhysioNet/CinC heart sound data.	Reported timing and segmentation metrics.	Segmentation is not source separation.	Requires PCG/ECG-style context.	Supports timing and evaluation considerations.
23	BUET multi-disease heart sound dataset: A comprehensive auscultation dataset for developing computer-aided diagnostic systems	Ali et al. (2026) (Ali et al., 2026)	Computer Methods and Programs in Biomedicine Update	Provide a public heart sound dataset.	Dataset descriptor.	BUET multi-disease heart sound recordings.	Provides labelled heart sound data for AI research.	Dataset is diagnosis-oriented, not paired separation data.	No lung source or mixed-source separation target.	Helps distinguish background datasets from separation datasets.
24	Cardiorespiratory Sound Separation Using Singular Spectrum Analysis	Han and Quan (2025) (Han and Quan, 2025)	International Conference on Signal Processing Systems	Separate cardiac and respiratory components.	Multi-stage singular spectrum analysis.	Digital stethoscope cardiorespiratory recordings.	Presents a recent direct separation approach.	Detailed evaluation evidence remains limited at this reporting stage.	Full implementation details need further confirmation.	Adds a recent non-deep-learning comparator.
25	StethAid: A Digital Auscultation Platform for Pediatrics	Arjoun et al. (2023) (Arjoun et al., 2023)	Sensors	Support pediatric digital auscultation workflow.	Digital auscultation platform.	Pediatric auscultation use context.	Demonstrates platform-oriented auscultation support.	Platform design is not separation-specific.	Does not provide heart-lung separation output.	Supports software workflow and interface thinking.
26	Sound-Dr: Reliable Sound Dataset and Baseline Artificial Intelligence System for Respiratory Illnesses	Hoang et al. (2023) (Hoang et al., 2023)	PHM Society Asia-Pacific Conference	Provide respiratory audio dataset and baseline AI.	Dataset and baseline AI system.	Sound-Dr respiratory recordings.	Supports respiratory sound dataset context.	Respiratory illness focus differs from source separation.	No paired heart-lung source targets.	Provides supplementary dataset background.
27	Compressed Sensing of Acoustic Cardiopulmonary Signals Using a	Baeyens et al. (2025) (Baeyens et al., 2025)	IEEE EMBC	Reconstruct acoustic cardiopulmonary signals.	Compressed sensing with CNN reconstruction.	Acoustic cardiopulmonary signals.	Shows reconstruction quality can affect later processing.	Reconstruction is not heart-lung source separation.	Dataset and deployment details need confirmation.	Supports acquisition and reconstruction context.

	CNN-based Reconstruction Method									
28	Real-Time Multi-Level Neonatal Heart and Lung Sound Quality Assessment for Telehealth Applications	Grooby et al. (2022) (Grooby et al., 2022)	IEEE Access	Assess neonatal heart and lung sound quality.	Real-time multi-level quality assessment.	Neonatal heart and lung sound recordings.	Supports input quality assessment before analysis.	Quality assessment is separate from source separation.	Does not generate separated outputs.	Supports input-quality checks for the prototype.
29	Bridging Auscultation and Tiny Machine Learning: A Digital Stethoscope Leveraging Convolutional Neural Networks on an Embedded Device for Organ Sound Analysis	Mutlu et al. (2024) (Mutlu et al., 2024)	Traitement du Signal	Process organ sounds on constrained devices.	CNN-based TinyML digital stethoscope system.	Organ sound recordings.	Demonstrates embedded auscultation processing.	Organ classification is not source separation.	Hardware direction is outside this software prototype.	Provides implementation context for practical systems.
30	Heart murmur detection from phonocardiogram recordings: The George B. Moody PhysioNet Challenge 2022	Reyna et al. (2023) (Reyna et al., 2023)	PLOS Digital Health	Benchmark heart murmur detection from PCG.	Challenge dataset and scoring framework.	CirCor DigiScope pediatric PCG data.	Provides benchmark context for heart sound analysis.	Detection task is not source separation.	Does not include lung source separation targets.	Supports dataset and evaluation background.
31	A robust deep learning based model for denoising phonocardiogram signals in clinical environments	Jakubec et al. (2025) (Jakubec et al., 2025)	Engineering Applications of Artificial Intelligence	Denoise PCG signals in noisy environments.	Deep learning denoising model.	PCG datasets.	Supports robust PCG noise reduction.	Denoising does not produce both heart and lung outputs.	Dataset and metric details require later confirmation.	Supports preprocessing design.
32	A self-attention-driven deep denoiser model for real time lung	Shuvo et al. (2026) (Shuvo et al., 2026)	Biomedical Signal Processing and Control	Denoise lung sounds in real time.	Self-attention deep denoiser.	Noisy lung sound recordings.	Shows attention-based	Lung denoising is not heart-lung separation.	Does not output separated cardiac sound.	Supports respiratory preprocessing choices.

	sound denoising in noisy environments						denoising potential.			
33	Adaptive thresholding of DWT coefficients using UNet for denoising real-life respiratory sounds	Behera et al. (2026) (Behera et al., 2026)	Biomedical Signal Processing and Control	Denoise real-life respiratory recordings.	DWT coefficient thresholding with UNet.	Real-life respiratory sounds.	Supports multiscale denoising for noisy respiratory data.	Focuses on respiratory denoising rather than source separation.	No paired heart-lung outputs.	Supports wavelet and UNet preprocessing ideas.
34	Lung sound signal denoising using discrete wavelet transform and artificial neural network	Pouyani et al. (2022) (Pouyani et al., 2022)	Biomedical Signal Processing and Control	Improve noisy lung sound quality.	DWT and artificial neural network.	Lung sound signals.	Supports classical and ML-based lung denoising.	Does not address cardiopulmonary source separation.	Dataset details need confirmation.	Supports preprocessing and denoising background.
35	Automatic breathing phase identification based on the second derivative of the recorded lung sounds	Pal et al. (2024) (Pal and Barney, 2024)	Biomedical Signal Processing and Control	Identify breathing phases from lung sounds.	Second-derivative based phase identification.	Heterogeneous lung sound recordings.	Reported breathing phase identification metrics.	Phase identification is not source separation.	Focuses only on respiratory signal structure.	Supports lung sound segmentation and evaluation context.

Appendix E: PRISMA Screening Summary

Below are the details of the PRISMA screening process used in Chapter 2.

Screening item	Count	Report interpretation
Records considered	96	Screened records relevant to cardiopulmonary sound separation and supporting literature
Records excluded	47	Excluded due to being out of scope, unavailable for supporting the current work, duplicates, or being irrelevant to the project
Eligible records after screening	49	Records left after the screening process and availability confirmation
Studies selected for Chapter 2 synthesis	35	Studies included in Chapter 2 synthesis, narrative, summary matrix, or full matrix
Core separation studies	18	Studies addressing cardiopulmonary, heart-lung, or chest separation
Background/supplementary studies	17	Studies addressing datasets, preprocessing, machine learning, evaluation, or other relevant topics
Eligible but unselected records	14	Records left as backup that were not used as evidence in the current paper

Appendix F: System Design Diagrams

The primary system design diagrams are presented in chapter 4. They include the prototype's context, use cases, activity, planned class structure, component-level architecture, and audio processing sequence. The diagrams are included to illustrate the design behind the prototype described in this thesis. All details not included in the diagrams were omitted for the sake of brevity.

Diagram	Purpose
Context diagram	The context diagram displays the relationships between the user, public repositories, and the proposed prototype.
Use case diagram	The use case diagram displays the cardiopulmonary sound separation system's use cases.
Activity diagram	The activity diagram displays the proposed system's sequence of activities from the request upload to download.
Class diagram	The class diagram displays the prototype's class structure and responsibilities.
Component diagram	The component diagram displays the browser interface, back-end API, service layer, model registry, separation strategies, database, and storage components.
Sequence diagram	The sequence diagram displays the processing sequence when the user uploads an audio file.

Appendix G: Planned Test Cases and Evaluation Metrics

The following planned tests are informed by the FYP2 implementation and evaluation. However, please note that only by implementing the prototype and conducting the tests, it will be possible to obtain results. In the table below, the test areas and the evidence for test completion are presented.

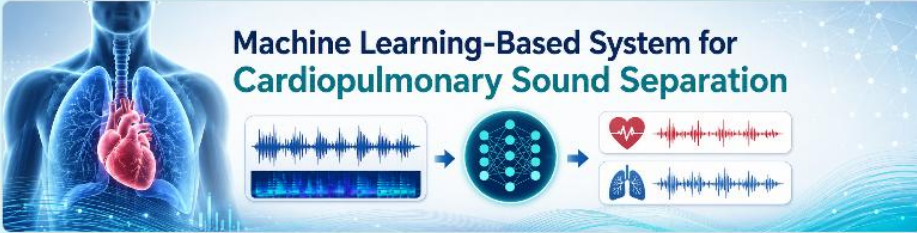
Test area	Planned check	Expected evidence
Audio upload	Accept a valid cardiopulmonary audio	Successful upload status and stored input metadata
File validation	Reject unsupported file type, missing file, or oversized file	Clear validation message not proceeding to the next step
Preprocessing	Convert audio into target sampling rate, channel format, and segment length	Stored preprocessing metadata and consistent model input
Model inference	Process a valid input of mixed audio with the selected method of separation	Heart and lung outputs
Output review	Provide the option to the user to interact, view, or download the separated outputs	Accessible final result view and downloadable .wav files
Metric calculation	Calculating evaluation metrics if reference sources are provided	An evaluation table with valid metric results
Error handling	Process a corrupted or incorrectly provided audio input without failure	User-informative error message and no erroneous output records

The table below includes the evaluation metrics that could be computed for cardiopulmonary prototypes if references sources were available.

Evaluation area	Planned metric or verification type	Purpose
Source separation quality	SDR, SI-SDR, SIR, or SAR if reference sources are available.	Estimate the degree of separation quality in relation to the target source.
Noise reduction	SNR or other signal quality indicator	Confirm that the preprocessing effectively reduced the level of noise
Audio reconstruction	Correlation, waveform inspection, or spectrogram comparison	Verify that the signal waveform was correctly reconstructed
System usability	Task completion and user feedback if user testing is approved.	Confirm that the system can be used to upload, process, and download the separated audio files.
Processing reliability	Planned test cases with success and failure conditions.	Verify that the prototype can consistently process both valid and invalid user inputs.

Appendix H: Full User Requirements Questionnaire

The following full set of questions was developed to establish traceability between the Google Form questions and the requirements analysis in chapter 3. In particular, although the requirements analysis in chapter 3 grouped the items under 10 broad requirement-focused categories for analysing and discussing the requirements, this appendix provides the full set of 24 questions as actually used in the Google Form survey for collecting the user requirements.



The diagram illustrates a machine learning-based system for cardiopulmonary sound separation. It features a human torso with a highlighted heart and lungs. To the right, a flowchart shows an input audio waveform being processed by a neural network (represented by a circle of nodes) to produce two separate output waveforms: one for the heart (with a heart icon) and one for the lungs (with a lung icon).

User Requirements Questionnaire for Machine Learning-Based Cardiopulmonary Sound Separation System

Welcome and thank you for participating in this questionnaire.

This questionnaire is conducted as part of a Final Year Project at **Multimedia University (MMU)** involving the development of a machine learning-based system for cardiopulmonary sound separation. The proposed system aims to allow users to upload a mixed cardiopulmonary audio recording and obtain separated heart and lung sound outputs.

The purpose of this questionnaire is to understand user background, current challenges in working with biomedical or audio-processing tasks, and user preferences regarding system features, interface design, output presentation, and evaluation needs.

Your responses will help identify user requirements and support the design of the proposed prototype. There are no right or wrong answers. All responses will be used for academic purposes only and will remain confidential.

Estimated completion time: 3-5 minutes.

Section A - Respondent Background:

Tell us about your background and experience with audio, machine learning, or biomedical signals.

1. What is your age group? *

- ☐ Under 18
- ☐ 18-25
- ☐ 26-35
- ☐ 36 and above

2. What best describes you? *

- ☐ Software Engineering / Computer Science student
- ☐ AI / Machine Learning student
- ☐ Biomedical Engineering/ Health-related student
- ☐ Researcher
- ☐ Lecturer / Supervisor
- ☐ Other:

3. How familiar are you with audio processing or signal processing? *

- ☐ Not familiar
- ☐ Basic
- ☐ Intermediate
- ☐ Advanced

4. Have you worked with biomedical audio before? *

- ☐ Yes
- ☐ No
- ☐ Not sure

5. Which type of audio-related task are you most familiar with? *

- ☐ Audio recording
- ☐ Audio editing
- ☐ Audio classification
- ☐ Audio separation
- ☐ Machine learning model testing
- ☐ I have not worked with audio tasks before

Section B - Current Challenges in Cardiopulmonary AudioProcessing:

Help us understand the main problems in separating heart and lung sounds from mixed audio.

1. How difficult do you think it is to separate heart and lung sounds from a mixed cardiopulmonary recording? *

- ☐ Very easy
- ☐ Easy
- ☐ Neutral
- ☐ Difficult
- ☐ Very difficult

2. What problems do you think affect cardiopulmonary sound processing the most? *

- ☐ Background noise
- ☐ Overlap between heart and lung sounds
- ☐ Low recording quality
- ☐ Difficulty identifying useful sound components
- ☐ Lack of simple software tools
- ☐ Lack of clear output visualization
- ☐ Other:

3. Do you think noisy audio recordings can reduce the quality of machine learning-based sound processing? *

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

4. Do you think a simple software prototype for heart-lung sound separation would be useful for students or researchers? *

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

Section C - System Feature Preferences:

Choose the features you think would make this sound separation system more useful.

1. Which feature would be most important in the proposed system? *

- ☐ Upload mixed cardiopulmonary audio
- ☐ Preprocess noisy audio
- ☐ Separate heart and lung sounds
- ☐ Preview separated audio
- ☐ Download separated outputs
- ☐ View processing history
- ☐ View evaluation results

2. Which output would you prefer after sound separation? *

- ☐ Heart and lung audio files only
- ☐ Audio files with waveform visualization
- ☐ Audio files with spectrogram visualization
- ☐ Audio files with evaluation metrics
- ☐ All of the above

3. How important is it for the system to provide a preview of the separated heart and lung sounds? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

4. How important is it for the system to allow users to download separated audio outputs? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

5. How important is it for the system to show basic processing information, such as file name, processing status, and completion time? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

6. Should the system include a history page showing previously uploaded files and separation results? *

- ☐ Yes
- ☐ No
- ☐ Not sure

Section D - Interface and Output Preferences:

Tell us how you want the system to show results, audio previews, charts, and messages.

1. How would you prefer the separated results to be displayed? *

- ☐ Simple audio player only
- ☐ Audio player with waveform
- ☐ Audio player with spectrogram
- ☐ Audio player with waveform, spectrogram, and metrics
- ☐ Not sure

2. Which interface style would you prefer? *

- ☐ Simple interface with only main features
- ☐ Detailed interface with many technical options
- ☐ Balanced interface with simple view and optional advanced details

3. How important is it for the system interface to be easy for non-expert users? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

4. What information should appear first after separation is completed? *

- ☐ Download buttons for heart and lung sounds
- ☐ Audio preview players
- ☐ Waveform or spectrogram visualization
- ☐ Evaluation metrics
- ☐ Processing status summary

5. What type of error message would be most helpful if the uploaded file is invalid? *

- ☐ Short message only
- ☐ Message explaining the problem
- ☐ Message explaining the problem and suggested fix
- ☐ Not sure

Section E - Evaluation and Improvement:

Share what would make the system more accurate, useful, and easier to improve.

1. How important is it for the system to show evaluation metrics for the separated output? *

- ☐ Not important
- ☐ Slightly important
- ☐ Moderately important
- ☐ Important
- ☐ Very important

2. Which evaluation information would be most useful? *

- ☐ Signal quality score
- ☐ Separation accuracy score
- ☐ Noise reduction information
- ☐ Processing time
- ☐ All of the above

3. Would you use a prototype that separates mixed cardiopulmonary audio into heart and lung sound outputs? *

- ☐ Yes
- ☐ No
- ☐ Maybe

4. What improvement or feature would you like to see in a cardiopulmonary sound separation system?

Long answer text

.....